THE FRESHWATER MUSSEL RESOURCES IN A SELECTED SEGMENT OF THE SALINE RIVER:

LOCATION, SPECIES COMPOSITION AND STATUS OF MUSSEL BEDS



Saline River Mussel Bed SRB70 Near Mt. Elba, Arkansas

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And
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THE FRESHWATER MUSSEL (BIVALVIA:UNIONACEA) RESOURCES IN A SELECTED SEGMENT OF THE SALINE RIVER: LOCATION, SPECIES COMPOSITION AND STATUS OF MUSSEL BEDS

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ABSTRACT

The decline of North American mussel communities has been attributed to several factors: anthropogenic perturbations including impoundments, sedimentation, dredging, and unrestricted land use; point source and non-point source contaminants, overharvesting by commercial shelling, and the introduction of exotic bivalve species. The mussel communities in Arkansas are no exception. Harris et al. (1997) designated 22 of the 75 (29%) freshwater mussel species native to Arkansas, to conservation status listing. Eight species are listed as federally endangered, and one is listed as Federally threatened.

The major objective of this study is to identify Federally listed threatened and endangered species of freshwater mussels, as well as global and state species of concern. Additional objectives included, documentation of species composition, approximate abundance and location of mussel beds. Approximately 73.5 km (45.7 river miles) of the Saline River were surveyed from near Tull, AR to AR Highway 167 between July 14 to September 22, 2001. In 2002 approximately 85.3 km (53 river miles) from AR Highway 167 to AR Highway 15 were surveyed between July 5 and September 9. Investigators searched suitable habitat by snorkel or "grubbing", characterized by feeling the substrate to find mussels and Hookah-rig, an oil-less air compressor with breathing line attached to a regulator.

A total of 95 mussel beds and 52 mussel concentrations was defined in the 164.2 km (101.8 rm) surveyed. Estimated average densities were generally 20 mussels/m^2 in defined areas upstream of Arkansas Highway 167. Estimated densities often exceeded 100 mussels/ m2 in beds downstream of Arkansas Highway 167, although estimated mean bed densities seldom exceeded 50 mussels/m^2. Estimated mean densities also were greater downstream of Arkansas Highway 167 ranging from 15 mussels/m^2 to 65 mussels/ m^2.

Dominant species within the survey area were *Elliptio dilatata* (27.4%) *Actinonaias ligamentina* (19.7%), *Fusconia flava* (8.4%), *Fusconia/Pleurobema* type (9.8%), *Pleurobema rubrum* (8.9%) and *Quadrula pustulosa* (6.7%), respectively. The federally listed species, *Lampsilis powelli*, *Lampsilis abrupta*, *Quadrula fragosa*, as well as 8 species of global and state concern, *Cyprogenia aberti*, *Lampsilis ornata*, *Ligumia recta*, *Toxolasma lividus*, *Obovaria jacksoniana*, *Villosa arkansasensis*, *Quadrula cylindrica*, and *Pleurobema rubrum* were encountered during this survey.

The total number of mussel beds for the mainstem of the Saline River is now 107 mussel beds, with 50.6 rm surveyed. This greatly exceeds the total number of mussel beds reported for any other river in Arkansas. Forty-two species are now known to occur in the Saline River, an additional 4 species may occur in backwater areas and genetic analyses may add as many as 4 additional species.

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INTRODUCTION

According to Williams et al. (1993) 72% of freshwater mussel (Bivalvia: Unionacea) species are considered extirpated, endangered, threatened or of special concern. The Nature Conservancy has recognized 55% of North American freshwater mussels as extinct or imperiled (Master 1990). Harris et al. (1997) designated 22 of the 75 (29%) freshwater mussel species native to Arkansas, to conservation status listing. Eight species are listed as federally endangered, and one is listed as federally threatened.

The decline of North American mussel communities has been attributed to several factors: anthropogenic perturbations including impoundments, sedimentation, dredging, and unrestricted land use; point source and non-point source contaminants, overharvesting by commercial shelling, and the introduction of exotic bivalve species. Habitat manipulation from impoundments created by dams, sedimentation, dredging, and unrestricted land use has resulted in changing life history function (Williams et al. 1993) and altering substrates preferred by mussels and host fish (Bates 1982). Decline in recruitment and regional die-offs have been attributed to historical as well as ongoing, unmonitored point and non-point sources (Neves 1997). The introduction of three exotic, invasive species of bivalve–the zebra mussel (*Dreissena polymorpha*), the dark false mussel (*Mytilopsis leucophaeata*) and the Asian clam (*Corbicula fluminea*)–within the rivers and lakes of North America have reduced available habitat, and food sources (Neves 1997). Additionally, *D. polymorpha* attaches to mussel valve margins, negatively affecting feeding, respiration, and burrowing to the point of extirpation (Neves 1997). All of these factors have the potential to affect mussel populations by disrupting their life cycle during any given stage (Neves 1997).

The Ouachita Valley, containing the Saline River, was formed by sequential events of glaciation, entrenchment and subsequent deposition of alluvial sediments including clays, silts, sands and gravels during the Quaternary period, approximately 2 million years ago (Saucier 1970). The quaternary period is comprised of the Pleistocene and the Holocene epochs. The Prairie and Deweyville terraces, recognized geologic features of the Saline River, are erosional landforms formed during these epochs. The Prairie has become less prominent due to erosion; however, the Deweyville terrace remains distinct. The Deweyville terraces were formed during glaciation and an increase in precipitation creating discharges 6 to 7 times that of present discharges (Saucier 1970, Saucier 1987).

During the Holocene period (past 12,000 to 14,000 years) to the present, the Saline River has maintained the same rate of flow and physical characteristics. The river has well defined meander belts and has seldom created new ones. The belts contain natural levees, many abandoned channels and depositional sand bars. Because the Saline River has rarely left its meander belts, the floodplain is now comprised of backswamp areas. (Saucier 1987).

Four tributaries of the Saline River – North, Middle, Alum and South forks – drain the eastern edge of the Ouachita Mountains. The tributaries converge with the main stem of the Saline just above the Gulf Coastal Plain ecoregion near Benton, Arkansas. The Saline

River then flows generally south, meandering through the Gulf Coastal Plain, and converges with the Ouachita River near the Arkansas-Louisiana state line. This reach of the river can be characterized as having low areas where swampy conditions persist with braided channels along its length. The riparian zone is dominated by bottomland hardwoods. The geology of the Ouachita Mountains is described as sandstone and shale and the Gulf Coastal Plain as sand, silts, clay and gravel. Slopes are generally 1% to 8%.

The drainage area for the Saline River at Arkansas Highway 15 near Rye, Arkansas is 2,102 mi^2 (USGS 1979). The mean flow at this location over the past 63 years is 2,601cfs. Recent land use along the Saline River has been predominately lumbering, combined with forest management practices for the production of pulpwood, poles and sawlogs (FTN 2002). According to a recent land use assessment (FTN 2002), 70% of land use is comprised of silviculture and approximately 8% and 6% pasture and cropland, respectively.

The Saline River is one of the last major undammed rivers in the Ouachita Mountain drainage. Arkansas Department of Environmental Quality (ADEQ) has designated the Saline River as an ecologically sensitive and extraordinary resource waterbody (ERW). The ecologically sensitive designation provides additional protection to waterbodies known to provide habitat for endangered, threatened or endemic species. The ERW designation protects a waterbody by recognizing its distinct combination of chemical, physical and biological attributes that is characterized by scenic beauty, aesthetics, scientific values, recreation potential and intangible social values.

ADEQ (2002) reported water quality conditions in the Saline River and its tributaries from data collected during 1998 - 2001. According to ADEQ the Saline River generally has a mean hardness of 38 mg $CaCO_3$/L, a mean pH of 6.9 and dissolved oxygen (D.O.) of 8 mg/L. Data indicate mineral content is elevated due to open pit bauxite mines and that small seasonal streams often have D.O. levels below 2 mg/L during the critical season in the Gulf Coastal Plain. A fish consumption advisory has been placed on 144.6 kilometers (km) (89.9 river miles; rm) of the Saline River because of mercury contamination. The source of contamination is derived from naturally occurring geological formations in the Ouachita Mountains and is manifested as runoff enters the slower lowland streams in the Gulf Coastal Plain (ADEQ 2002). The Saline and its many tributaries are receive effluent waters from 49 facilities with National Pollution Discharge Elimination System (NPDES) permits issued by ADEQ. Sixteen of these permits have been issued to city or town wastewater treatment facilities and the remainder are commercial or schools.

Gordon et al. (1979) reviewed literature and museum specimens on Arkansas' unionacean fauna and subsequently listed 35 species as known or possible elements of the Saline River unionacean fauna. Headwater reaches of the Saline River to the Saline/Grant County line, Arkansas have been adequately surveyed (Harris and Gordon 1988; Brown and Brown 1989; U.S. Fish and Wildlife Service 1990; U.S. Fish and Wildlife Service 1992; Burns & McDonnell 1992a, 1992b; Davidson and Gosse 2001). Davidson (1997) surveyed the lower 18 km of the Saline River that lies within the

boundary of Felsenthal National Wildlife Refuge. A few unionid records exist for highway crossings (John Harris, AHTD) on the Saline River. Otherwise, species composition, abundance and mussel bed locations are unknown for a 238.1 km (148.0 rm) section of the Saline River from near Tull, Arkansas to the northern boundary of Felsenthal National Wildlife Refuge.

Objectives

The major objective of this study was to identify Federally listed threatened and endangered species of freshwater mussels, as well as global and state species of concern. Additional objectives include, documentation of species composition, approximate abundance and location of mussel beds. Results will provide baseline information on the status of the mussel fauna inhabiting the Saline River. The data derived from this survey will include: mussel bed dimensions, water depth, habitat type, mapped locations of mussel beds, any apparent stressors; and biological data such as species composition, and approximate abundance.

METHODS AND PROCEDURES

The Saline River from near Tull to Canoe Landing was traversed upstream to downstream using a canoe to locate mussel concentrations and beds. Reaches from Arkansas Highway 167 to Arkansas Highwy 15 on the Saline River were accessed using a motorized flat bottom boat. Approximately 73.5 km (45.7 rm) were surveyed from near Tull, Arkansas to Arkansas Highway 167 during July 14 to September 22, 2001. Approximately 85.3 km (53 rm) from Arkansas Highway 167 to Arkansas Highway 15 were surveyed during July 5 to September 9, 2002 (Figure 1).

Investigators searched favorable habitat and/or approximately 500 to 600 foot intervals in deeper water for mussels. Shallow areas, riffles and runs were searched by snorkel or "grubbing", characterized by feeling the substrate to find mussels. These concentrations and beds were searched for approximately 2/3 man-hour (m-h). In deeper water, mussel concentrations and beds were searched by Hookah-rig, an oil-less air compressor with breathing line attached to a regulator, for 1.25 m-h. Mussel beds are defined as areas $\geq$100 m^2 with mean mussel densities $\geq$10 mussels/m^2. Mussel concentrations are defined as areas with mussels covering <100 m^2 and/or with mean mussel densities <10 mussels/m^2.

Mussels were handpicked and placed in a bag and brought to the surface, identified and enumerated. Nomenclature follows Turgeon and others (1998). Mussel bed or concentration dimensions, location, densities, water depth, habitat/substrate type were recorded, as well as any apparent stressors (i.e. sloughing banks, point or non-point source impacts). Bed length was determined to the nearest meter using a Bushnell Yardage Pro laser rangefinder. Bed and concentration locations were marked on the appropriate (U.S. Geological Survey) topographic map. Water depth was estimated in shallow areas and measured to the nearest foot with a Lawrance electronic depth finder in deeper water. Three voucher specimens of each species not considered state or Federal

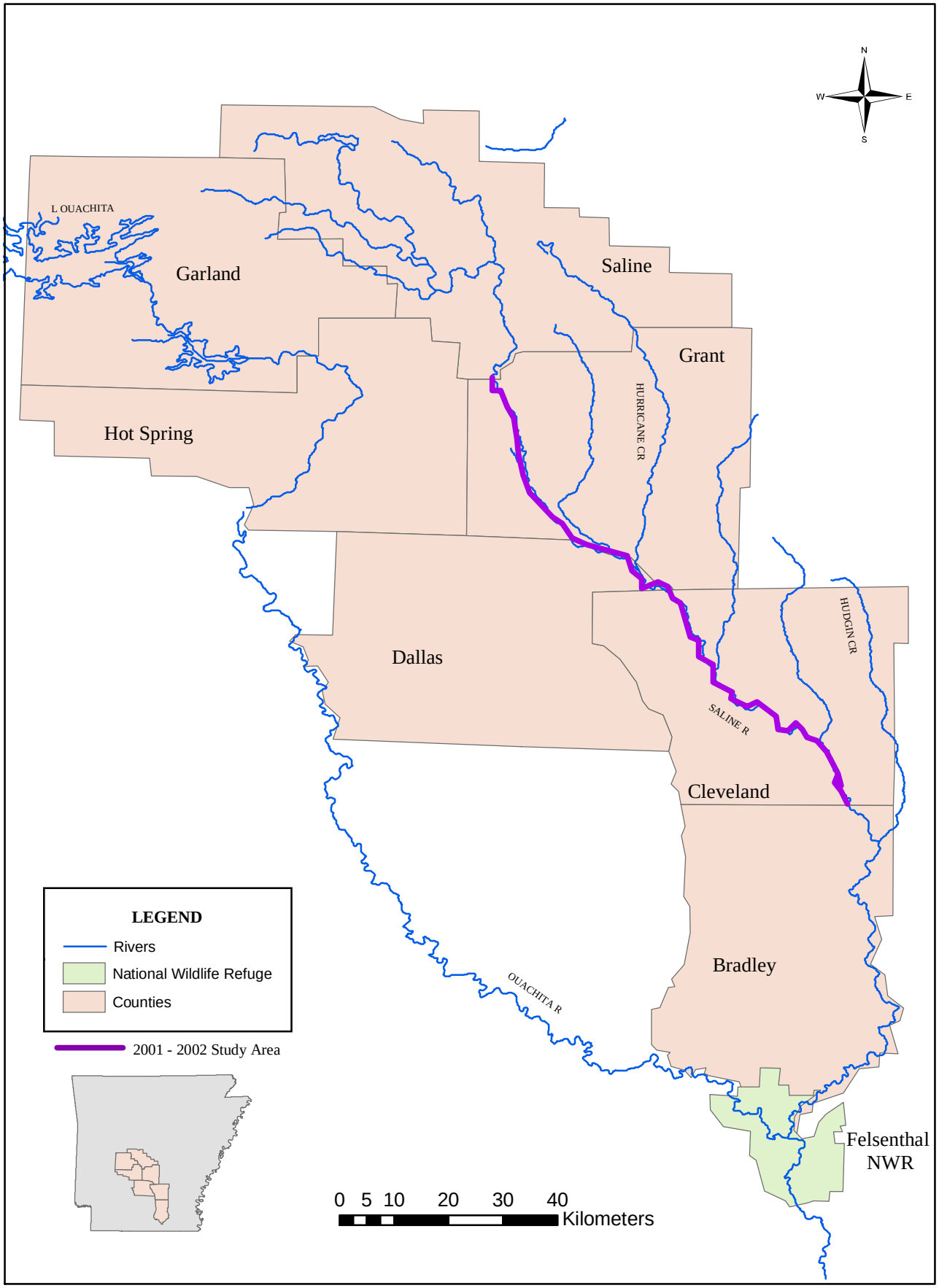


Figure 1. Saline River study area from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

Endangered and Threatened were donated to the Arkansas State University Museum of Zoology, Unionacea Collection. "Listed" species were photographed to verify identification.

RESULTS

Mussel beds and concentrations were generally located in three physical settings: 1) riffle/run areas; 2) along descending banks immediately upstream of riffle/run areas; or 3) in river bends. Mussels were located in water depths, at low flow conditions, ranging from approximately 0.2 m to 4 m. Mussels were most often associated with gravel/sand. In some areas upstream of Arkansas Highway 167, dense mats of algae covered the gravel. Mussels in these areas appeared to be migrating upward from the gravel/sand substrate into algal mats, which acted as sediment traps. Pools corresponding with long straight stretches generally were devoid of mussels.

A total of 95 mussel beds and 52 mussel concentrations was defined in the 164.2 km (101.8 rm) surveyed (Appendix 1; Figure 2). Addendum one, a series of U. S. Geological Survey topographic maps with mussel beds and concentrations marked and labeled, was provided to each funding agency. Forty-one species were found within the study area (Table 1). Of these, three species are listed by the U. S. Fish and Wildlife Service (USFWS) as Federally Threatened or Endangered, seven species are listed by The Nature Conservancy (TNC) as G1 or G2 and 11 species are listed by the Arkansas Natural Heritage Commission (ANHC) as S1 or S2. Table 1 summarizes the conservation status by the USFWS, TNC and ANHC (ANHC 2002).

Estimated average densities were generally < 20 mussels/m^2 in defined areas upstream of Arkansas Highway 167. Estimated densities often exceeded 100 mussels/m^2 in beds downstream of Arkansas Highway 167, although estimated mean bed densities seldom exceeded 50 mussels/m^2. Estimated mean densities also were greater downstream of Arkansas Highway 167 ranging from 15 mussels/m^2 to 65 mussels/m^2. In the stream reach from near Tull, Arkansas to Arkansas Highway 167, 2 of 28 mussel beds and 2 of 34 mussel concentrations were > 500 m^2, but < 1,000 m^2. Two beds > 500 m^2, but < 1,000 m^2, were identified between Arkansas Highway 167 and SRB51 (approximately 1 rm upstream of Canoe Landing). Twenty-seven mussel beds > 500 m^2 were identified from SRB52 (Canoe Landing) to Arkansas Highway 15. Of these, 10 mussel beds were 500 to 1,000 m^2, 17 mussel beds were 1,000 to 5,000 m^2 and 2 mussel beds were > 5,000 m^2. The largest mussel bed was 8,000 m^2. No mussel concentrations exceeded 500 m^2 downstream of Arkansas Highway 167.

Community Composition

A total of 11,204 individuals were examined. *Elliptio dilatata* (27.4%) was the dominant species within the survey area. Other common species included *Actinonaias ligamentina* (19.7%), *Fusconaia flava* (8.4%), *Fusconaia/Pleurobema* type (9.8%), *Pleurobema rubrum* (8.9%) and *Quadrula pustulosa* (6.7%). Federally Threatened and Endangered

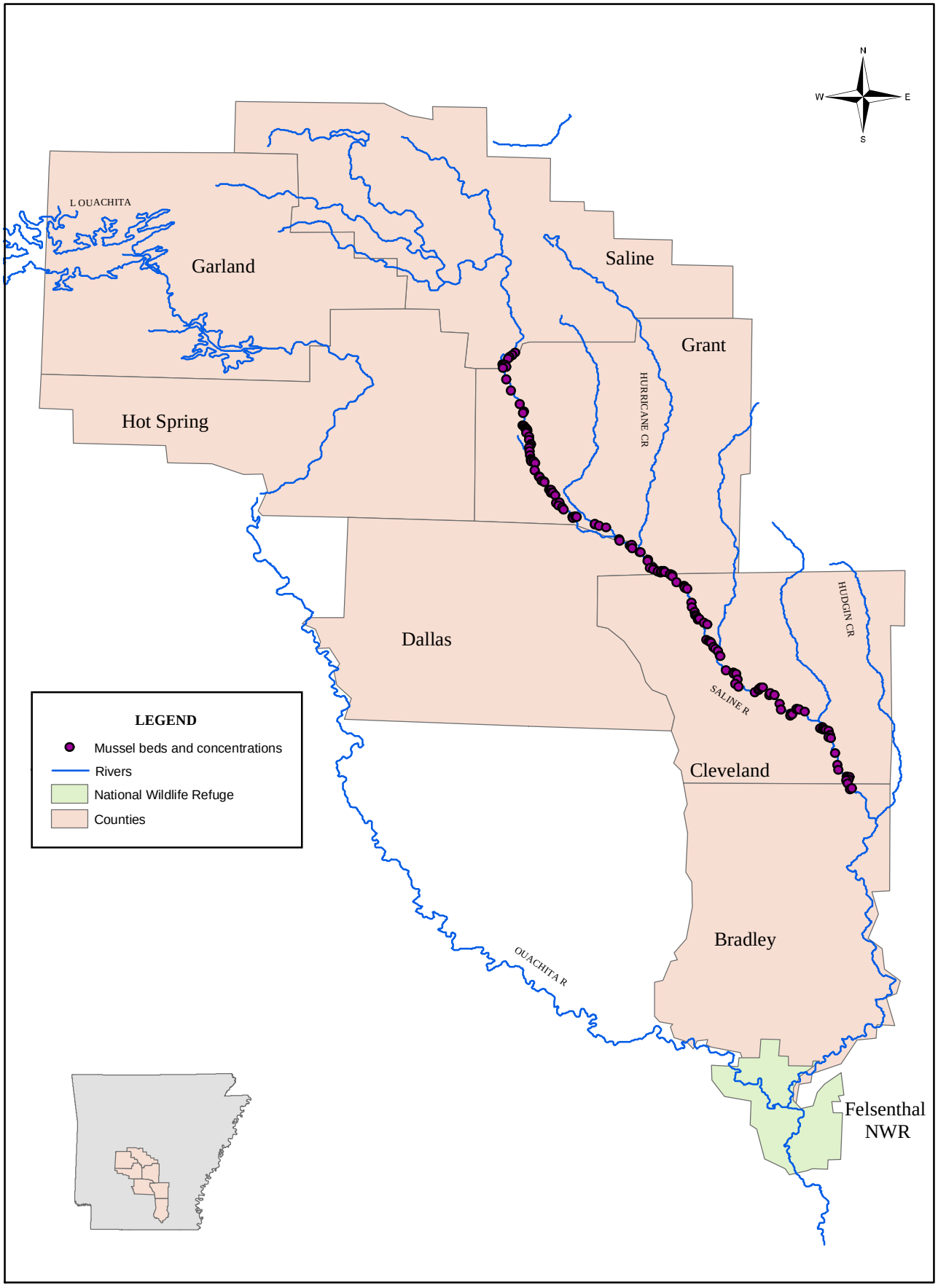


Figure 2. Freshwater mussel (Unionacea) bed and concentration locations in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

Table 1. Freshwater mussel (Unionacea) fauna occurring in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas.

Species	Federal Status	Global Rank	State Rank
Actinonaias ligamentina	-	G5	S5
Alasmidonta marginata	-	G4	S3
Amblema plicata	-	G5	S5
Cyprogenia aberti	-	G2	S2
Ellipsaria lineolata	-	G4	S3
Elliptio dilatata	-	G5	S4
Fusconaia ebena	-	G4G5	S3S4
Fusconaia flava	-	G5	S4
Fusconaia/Pleurobema type	-	-	-
Lampsilis abrupta	E	G2	S2
Lampsilis cardium	-	G5	S4
Lampsilis ornata	-	G5	S1
Lampsilis hydiana	-	G4	S3
Lampsilis powelli	T	G1G2	S2
Lampsilis teres	-	G5	S4
Lasmigona costata	-	G5	S3
Leptodea fragilis	-	G5	S4
Ligumia recta	-	G5	S2
Megalonaias nervosa	-	G5	S3
Obliquaria reflexa	-	G5	S4
Obovaria jacksoniana	-	G1G2	S2
Obovaria olivaria	-	G4	S3
Plectomerus dombeyanus	-	G4	S4
Pleurobema rubrum	-	G2	S2
Potamilus purpuratus	-	G5	S4
Ptychobranchus occidentalis	-	G3G4	S3
Pyganodon grandis	-	G5	S5
Quadrula fragosa	E	G1	S1
Quadrula cylindrica	-	G3	S2
Quadrula metanevra	-	G4	S3S4
Quadrula nodulata	-	G4	S4
Quadrula pustulosa	-	G5	S5
Quadrula quadrula	-	G5	S5
Strophitus undulates	-	G5	S3
Toxolasmus lividus	-	G2	S2
Tritogonia verrucosa	-	G4	S4
Truncilla donaciformis	-	G5	S3
Truncilla truncata	-	G5	S4
Utterbackii imbecillis	-	G5	S3S4
Villosa arkansasensis	-	G2	S2
Villosa lienosa	-	G5	S3

S1 – Critically imperiled in the state because of extreme rarity or because of some factor(s) making it especially vulnerable to extirpation from the state. (Typically 5 or fewer occurrences or very few remaining individuals)
S2 – Imperiled in the state because of rarity or because of some factor(s) making it very vulnerable to extirpation from the state. (6 to 20 occurrences or few remaining individuals)
S3 – Rare and uncommon in the state. (21 to 100 occurrences)
S4 – Widespread, abundant, and apparently secure in state, with many occurrences, but the element is of long-term concern. (Usually more than 100 occurrences)
S5 – Demonstrably widespread, abundant, and secure in the state, and essentially ineradicable under present conditions.
(ANHC 2002)

Global rankings are reported by The Nature Conservancy.

T = Threatened; E = Endangered

Table 2. Species Composition, Saline River from near Tull, AR to AR Highway 15 nearWarren, AR, 2001-2002.

Scientific Name	Tot Ind.	% Tot
Actinonaias ligamentina	2,210	19.73
Alasmidonta marginata	7	0.06
Amblema plicata	391	3.49
Cyprogenia aberti	254	2.27
Ellipsaria lineolata	13	0.12
Elliptio dilatata	3,066	27.37
Fusconaia ebena	108	0.96
Fusconaia flava	942	8.41
Fusconaia/Pleurobema type	1,093	9.76
Lampsilis abrupta	14	0.12
Lampsilis cardium	170	1.52
Lampsilis hydiana	2	0.02
Lampsils powelli	18	0.16
Lampsilis teres	14	0.12
Lasmigona costata	8	0.07
Leptodea fragilis	4	0.04
Ligumia recta	39	0.35
Megalonaias nervosa	14	0.12
Obliquaria reflexa	28	0.25
Obovaria jacksoniana	34	0.30
Obovaria olivaria	1	0.01
Plectomerus dombeyanus	76	0.68
Pleurobema rubrum	1,001	8.93
Potamilus purpuratus	16	0.14
Ptychobranchus occidentalis	389	3.47
*Pyganodon grandis**	0	0.00
Quadrula cylindrica	6	0.05
Quadrula fragosa	1	0.01
Quadrula metanevra	332	2.96
Quadrula nodulata	13	0.12
Quadrula pustulosa	753	6.72
Quadrula quadrula	2	0.02
Strophitus undulatus	46	0.41
Toxolasma lividus	4	0.04
Tritogonia verrucosa	66	0.59
Truncilla donaciformis	3	0.03
Truncilla truncata	61	0.54
*Utterbackii imbecillis**	0	0.00
Villosa arkansasensis	2	0.02
Villosa lienosa	3	0.03
Number of Live Individuals	11,204	100.00

* Represents taxa collected outside mussel beds and concentrations.

species, TNC G1 and G2 species and ANHC S1 and S2 species accounted for 0.3%, 11.9% and 12.3% of the total community, respectively (Table 2). Excluding the fourth most common species *P. rubrum* in the survey area, TNC G1 and G2 species and ANHC S1 and S2 species comprise 3.0% and 3.5% of the total community, respectively.

Longitudinal Distribution

Actinonaias ligamentina was (co) dominant in 16 of 28 mussel beds and concentrations upstream of RMR Camp, located approximately 10.5 km (6.5 rm) downstream of Arkansas Highway 270 (Appendix 2; Addendum 1). Other (co) dominant species in this reach include *Fusconaia flava* (5), *Ptychobranchus occidentalis* (3), *Elliptio dilatata* (2), *Fusconaia flava* (1), *Fusconaia ebena* (1), *Lampsilis cardium* (1), *Toxolasma lividus* (1) and *Quadrula pustulosa* (1).

Fusconaia/Pleurobema type was (co) dominant in 20 of 34 mussel beds and concentrations between RMR Camp and Arkansas Highway 167 (Appendix 2; Addendum 1). Other (co) dominant species in this reach include *Pleurobema rubrum* (8), *Elliptio dilatata* (3), *Actinonaias ligamentina* (3), *Fusconaia flava* (2), *Plectomerus dombeyanus* (1) and *Quadrula pustulosa* (1).

Elliptio dilatata was (co) dominant in 56 of 68 mussel beds and concentrations between Arkansas Highway 167 and SRB79, located approximately 3.2 km (2 rm) upstream of Vince Bluff (Appendix 2; Addendum 1). Other dominant species included *Actinonaias ligamentina* (7), *Fusconaia/Pleurobema* type (4), *Fusconaia flava* (1) and *Pleurobema rubrum* (1). To help better analyze distributional trends, *E. dilatata* was (co) dominant in 42 of 44 mussel beds and concentrations from Mahoney's Ferry (SRB43) to SRB79. *A. ligamentina* relative abundance increased in the reach from SRB80 to Arkansas Highway 15. In this reach, *A. ligamentina* and *E. dilatata* dominated 8 of 17 and 9 of 17 mussel beds,

respectively. *Amblema plicata* and *Quadrula pustulosa* comprise a greater portion of bed composition downstream of Mt. Elba.

Federally Threatened and Endangered Species

Lampsilis powelli (Lea 1852) – Arkansas Fatmucket. Distribution: Figure 3. Status: Federal – Threatened, TNC – G1/G2, State – S2. Plate 1.

L. powelli is restricted to portions of the Saline River upstream of Arkansas Highway 270. *L. powelli* was found in moderate to swift currents over sand with gravel and gravel substrates. Riffle/run sequences generally provided the best habitat. Nine sites were identified, including one relict site (SRC2; SRC5; SRB1; SRB2; SRC6; SRC8; SRB4; SRC14; SRC20). This extends the known range of *L. powelli* by 41.9 km (26 rm). SRB2 was the only site with multiple individuals (11) collected, which were 5.8% of the mussels examined in SRB2. *L. powelli* population numbers appear stable upstream of Arkansas Highway 270, but low numbers occur in the periphery of this species range.



Plate 1. *Lampsilis powelli*

Lampsilis abrupta (Say 1831) – Pink Mucket. Distribution: Figure 4. Status: Federal – Endangered, TNC – G2, State – S2. Plates 2 and 3.

L. abrupta was found throughout the survey area. Inside banks comprised of sand with gravel substrate generally provided the most suitable habitat. This species is considered extremely rare upstream of Arkansas Highway 167, with only 2 sites identified (SRB4; SRC21). *L. abrupta* was more common downstream of Arkansas Highway 167, with 11 sites identified (SRB27; SRC36; SRB31; SRB40; SRB43; SRB45; SRC49; SRB74; SRB78; SRB82; SRB85). The maximum number of individuals sampled from a bed or concentration was two (SRB85), which were 1.2% of individuals examined from SRB85. *L. abrupta* population numbers appear unstable due to low numbers and limited dispersal in the Saline River. One individual, an apparent juvenile (4 to 7 years based on external annuli), was collected from SRB40. This may suggest that limited reproduction is occurring within the population.



Plate 2. *L. abrupta*



Plate 3. *L. abrupta* site (SRB78)

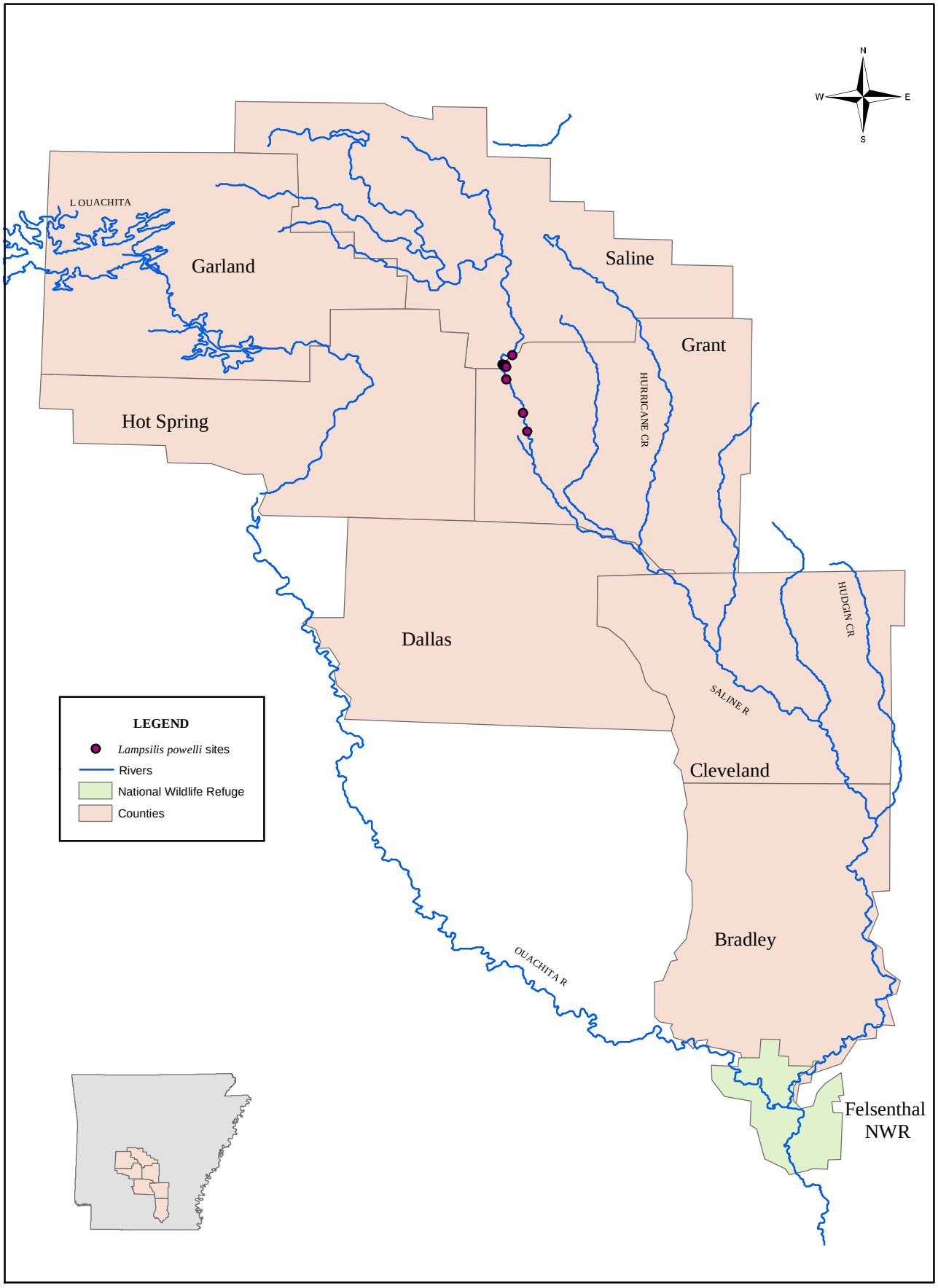


Figure 3. *Lampsilis powelli* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

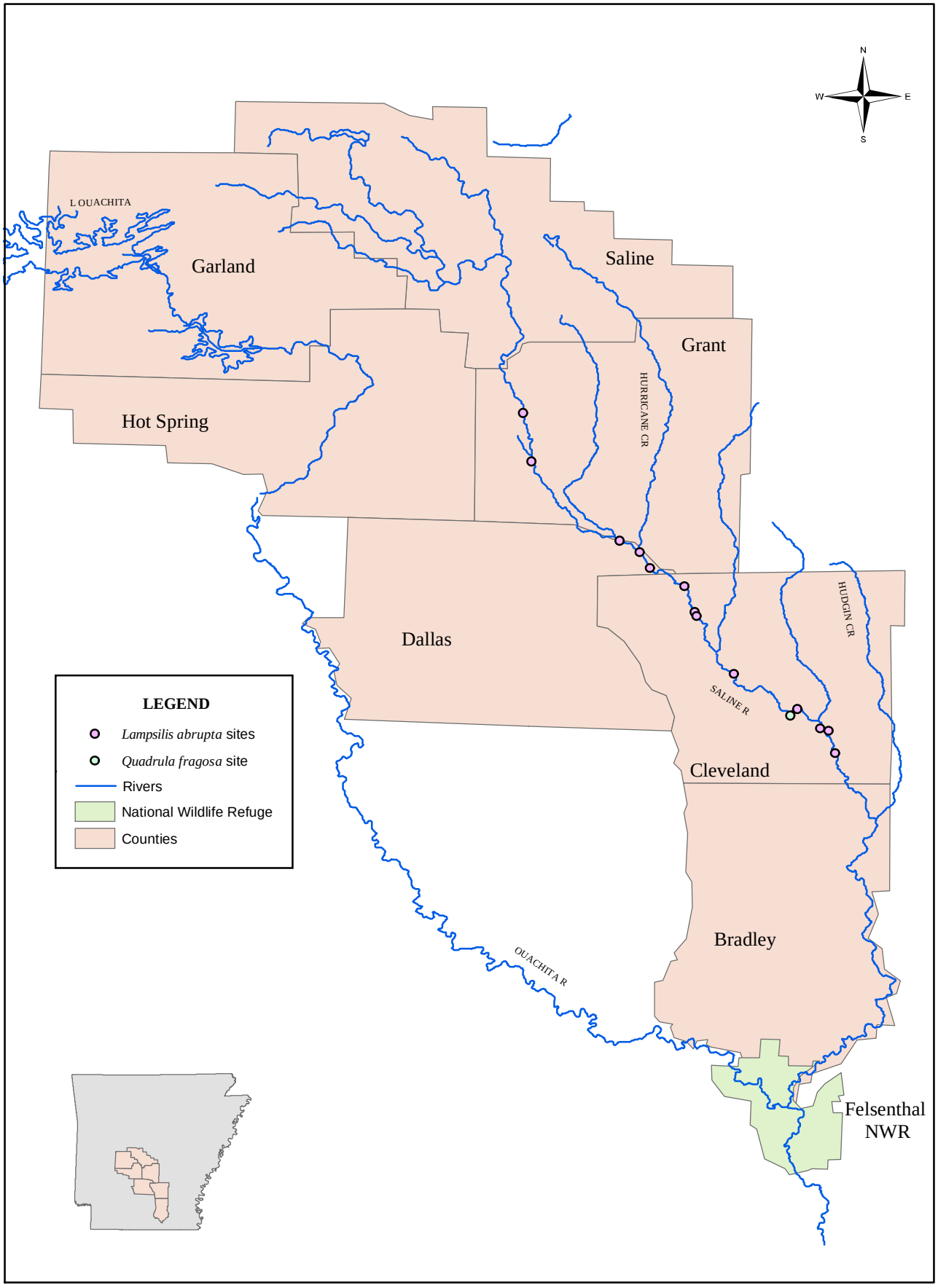


Figure 4. *Lampsilis abrupta* and *Quadrula fragosa* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

Quadrula fragosa (Conrad 1835) – Winged Mapleleaf. Distribution: Figure 4. Status: Federal – Endangered, TNC – G2, State – S1. Plates 4 and 5.

Q. fragosa was collected from one site (SRB71) approximately 3.2 km (2 rm) downstream of Mt. Elba. The substrate consisted of gravel with sand. This represents only the 4th known location and 8th individual reported for this species in Arkansas. The other three locations include two sites in the Ouachita River upstream of Camden, Arkansas and 1 site in the Little Missouri River near its confluence with the Ouachita River. This also represents the first location in the Ouachita River Basin that is a substantial distance (approximately 109.7 km; 68 rm) from the mainstem Ouachita River. *Q. fragosa* population numbers are at risk of being extirpated from this portion of the Saline River.



Plate 4. *Q. fragosa*



Plate 5. *Q. fragosa* site (SRB71)

Species of Global and State Concern

Cyprogenia aberti (Conrad 1850) – Western Fanshell. Distribution: Figure 5. Status: TNC – G2, State – S2. Plate 6.

C. aberti is common (10th most common species) throughout the survey area, although it is less common upstream of Arkansas Highway 270 (4 live individuals from 2 sites). Two hundred fifty-four (254) individuals representing 2.3% of the total community were collected from 89 sites. *C. aberti* generally occurs in low numbers, but are locally abundant in the reach from Mt. Elba to Arkansas Highway 15. Ongoing genetic research may reveal that



Plate 6. *C. aberti*

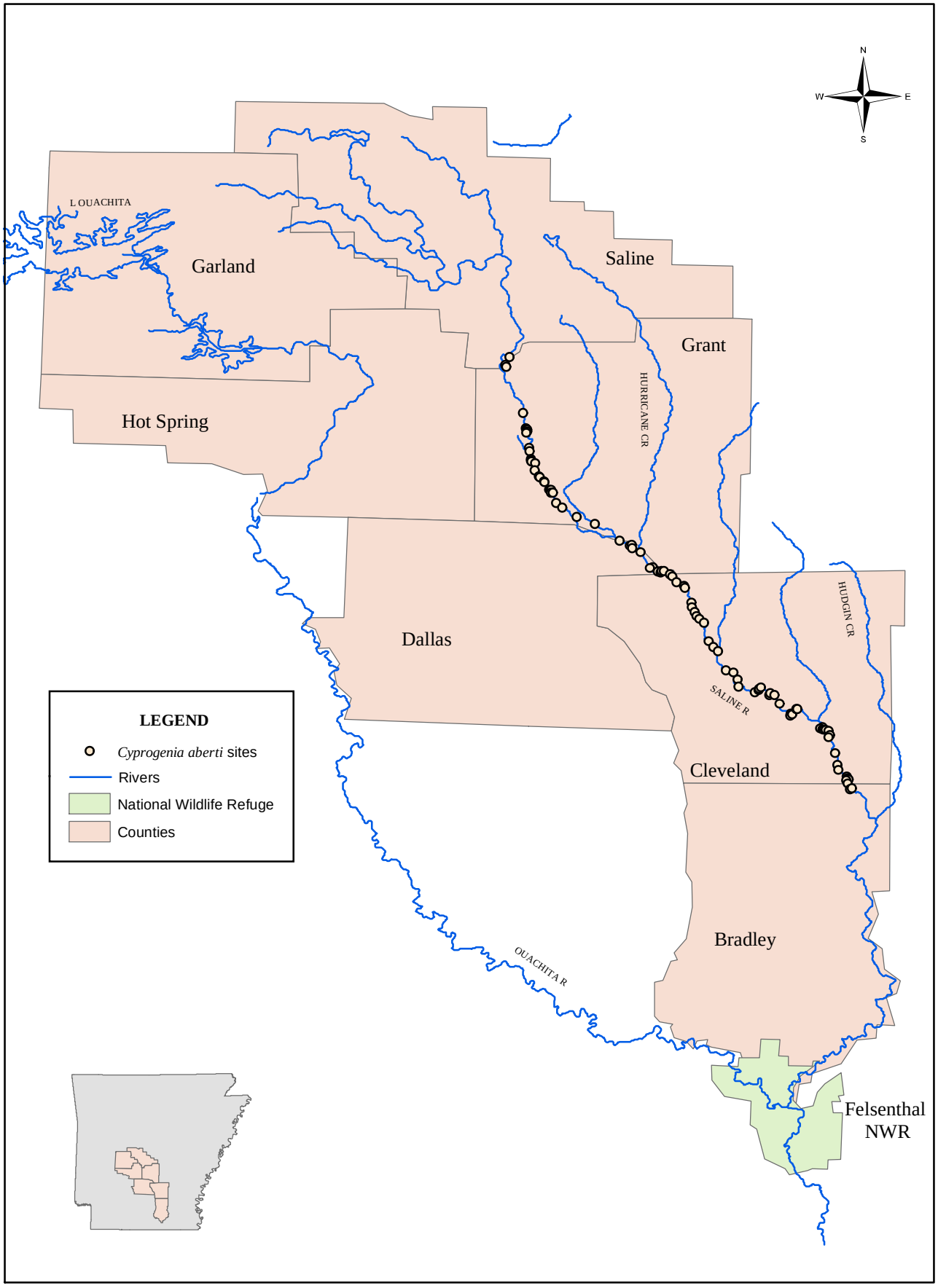


Figure 5. *Cyprogenia aberti* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

Another *Cyprogenia* species occurs within the same type localities as *C. aberti*. For this reason, *C. aberti* specimens were given to John Harris for genetic analysis. At the time of this report, *C. aberti* is the only recognized *Cyprogenia* species that occurs in Arkansas. All individuals collected from the Saline River during this survey were considered *C. aberti*, although future studies may discover the coexistence of another *Cyprogenia* species with *C. aberti* in the Saline River.

Lampsilis ornata (Conrad 1835) – Southern Pocketbook. Status: TNC – G5, State – S1.

L. ornata was recently recognized as occurring in Arkansas. Davidson and Gosse are preparing a manuscript for publication on the status and distribution of mussels inhabiting a portion of the Saline River near the confluence of Holly Creek (upstream of Tull, Arkansas), and *L. ornata* was discovered in this area of the river (Davidson and Gosse 2003). Since information regarding the presence of *L. ornata* in Arkansas was not available during the first half of this survey (upstream of Arkansas Highway 167), it is very probable that some specimens considered *L. cardium* may have been *L. ornata.* However, no specimens of *L. ornata* were encountered downstream of Arkansas Highway 167. The authors believe that *L. ornata* is restricted to the same reaches as *L. powelli* (upstream of Arkansas Highway 270). For these reasons, we have included *L. ornata* in the taxa list for the Saline River.

Ligumia recta (Lamarck 1819) – Black Sandshell. Distribution: Figure 6. Status: TNC – G5, State – S2. Plate 7.

L. recta occurred throughout the survey area. However, it was restricted to two sites (SRB37; SRB40) between SRC26, near Jenkins Ferry State Park, and SRB59, near Arkansas Highway 79. No apparent reasons were obvious for its near absence from this portion of the Saline River. This species appears to be stable in reaches upstream and downstream of the above mentioned segment, generally occurring in low numbers. Apparent juvenile individuals were discovered at multiple sites suggesting that reproduction is occurring within this population.



Plate 7. *L. recta*

Toxolasma lividus (Rafinesque 1831) – Purple Liliput. Distribution: Figure 7. Status: TNC – G2, State – S2.

T. lividus was discovered at 6 sites in the survey area, of which three were relict sites. The authors believe that this species may be more common in muddy areas in the Saline River. These areas typically do not support large numbers of mussels or diverse assemblages and typically were not surveyed. This species

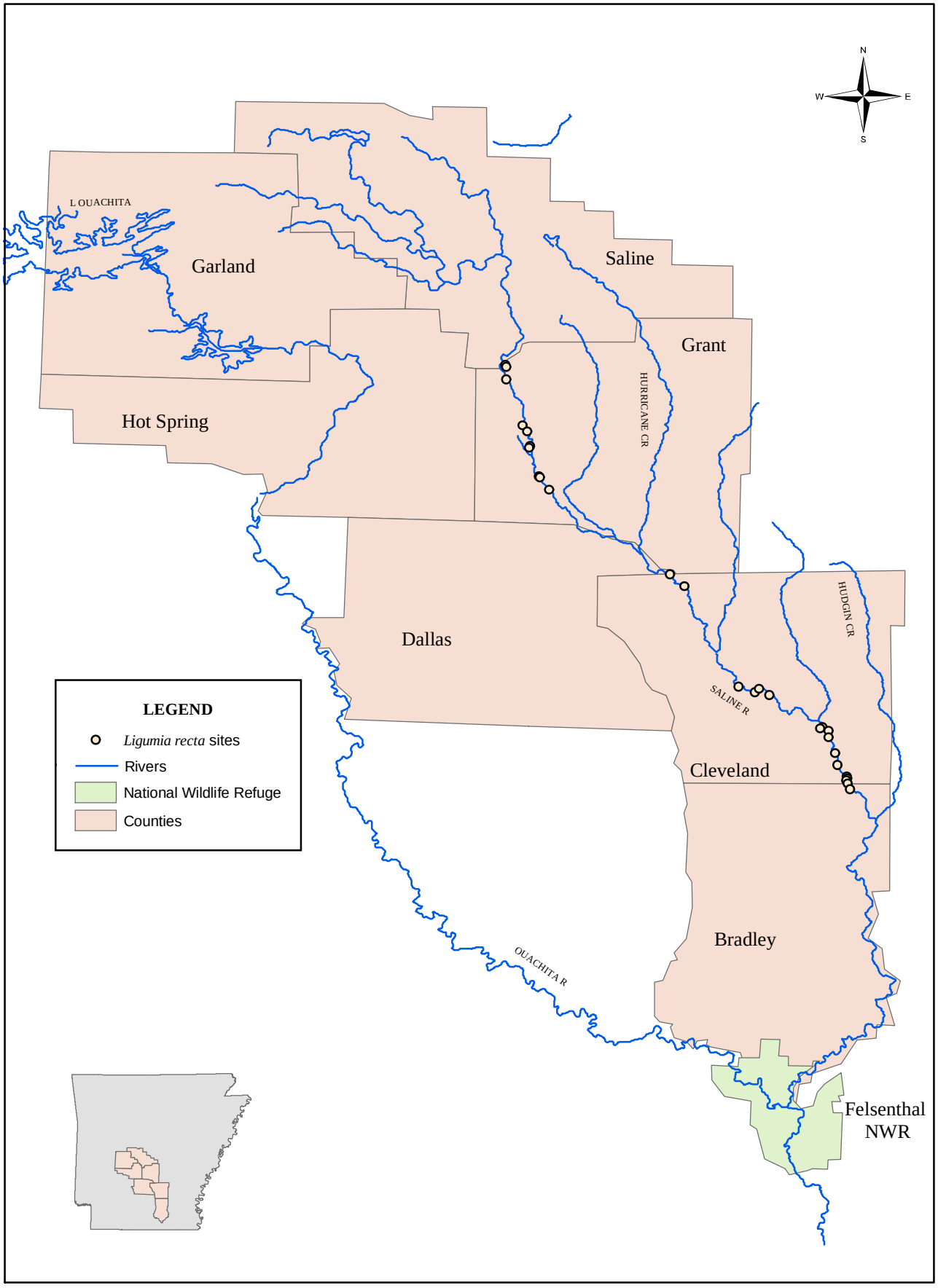


Figure 6. *Ligumia recta* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

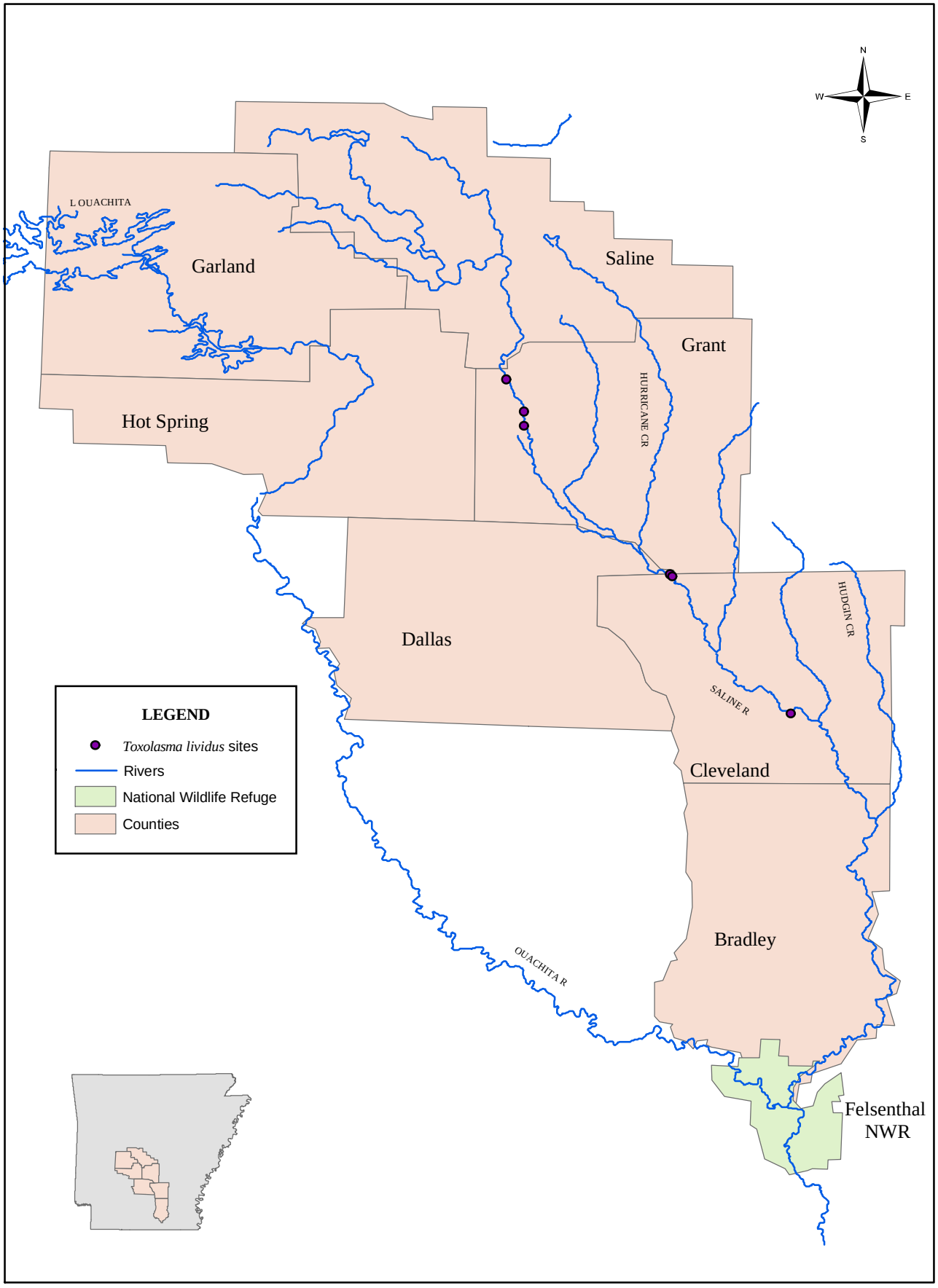


Figure 7. *Toxolasma lividus* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

occurred throughout the survey area in very low numbers. Numerous (40+) fresh dead individuals were discovered in SRC10. This site was located along a descending bank in a large pool and represented the greatest concentration of *T. lividus*.

Obovaria jacksoniana (Frierson 1912) – Southern Hickorynut. Distribution: Figure 8. Status: TNC – G1/G2, State – S2.

O. jacksoniana occurs (23 sites) throughout the survey area. This species is rare upstream of Mt. Elba, becoming increasingly more common downstream of Mt. Elba. Multiple individuals were collected from one site upstream of Mt. Elba and five sites downstream of Mt. Elba (Appendix 2). *O. jacksoniana* is widely distributed in the Saline River, but nowhere abundant.

Villosa arkansasensis (Lea 1862) – Ouachita Creekshell. Distribution: Figure 9. Status: TNC – G2, State – S2

V. arkansasensis is considered a headwater species and was discovered at two sites (SRC20; SRC34), represented by one individual per site, in the survey area (Appendix 2). This species is restricted to reaches upstream of Arkansas Highway 167. *V. arkansasensis* population numbers appear very unstable, in the survey area, because of its rarity.

Quadrula cylindrica (Say 1917) – Rabbitsfoot. Distribution: Figure 10. Status: TNC – G3, State – S2. Plate 8.

Q. cylindrica was restricted to 3 sites (SRB69; SRB70; SRB72) near Mt. Elba, in a two-mile segment of the Saline River. SRB70 had the greatest number of individuals (4) collected, which was 3.0% of the total community. One individual was collected from each of the other two sites. Since this species is restricted to a two-mile reach and occurs in low numbers, the authors feel that *Q. cylindrica* population numbers are unstable and one disturbance could extirpate this species from the survey area.



Plate 8. *Q. cylindrica*

Pleurobema rubrum (Rafinesque 1820) – Pyramid pigtoe. Distribution: Figure 11. Status: TNC – G2, State – S2.

P. rubrum was widespread throughout the study area. This species was the fourth most common (8.9%) species encountered in the survey area. *P. rubrum* dominated ten beds or concentrations, most (8) occurring between RMR Camp and Arkansas Highway 167. This species appears to be stable within the survey area, often occurring at high densities.

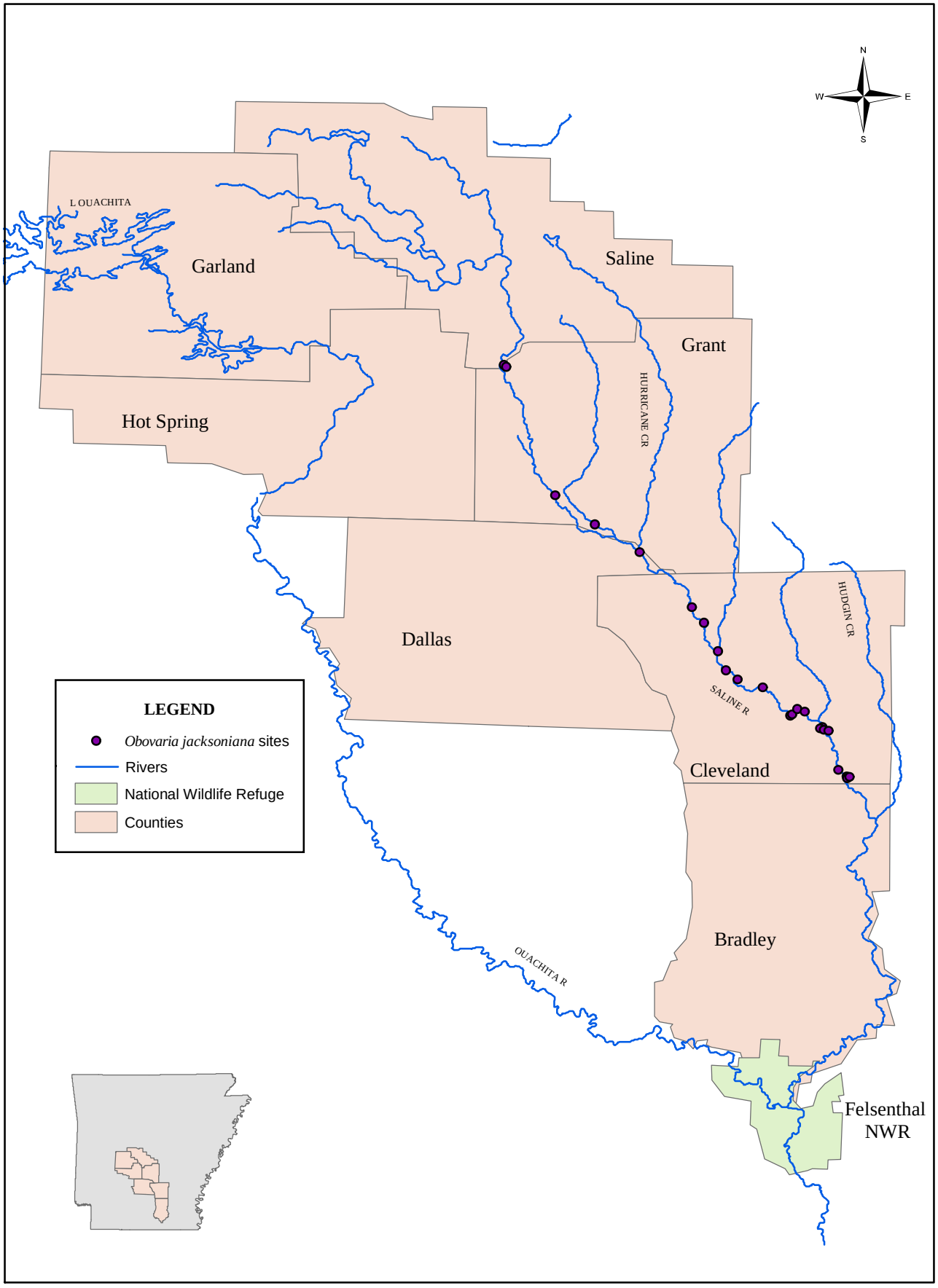


Figure 8. *Obovaria jacksoniana* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

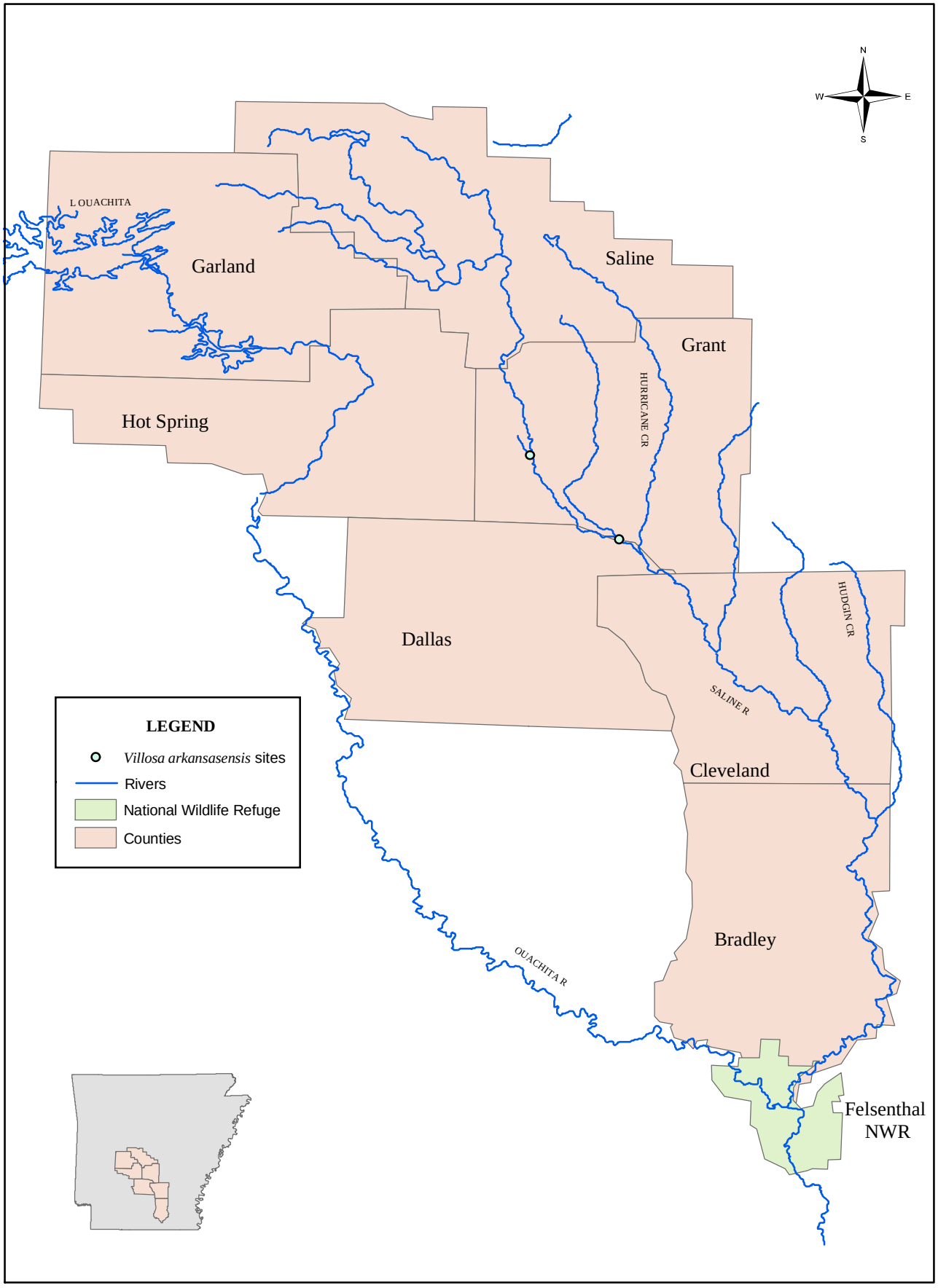


Figure 9. *Villosa arkansasensis* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

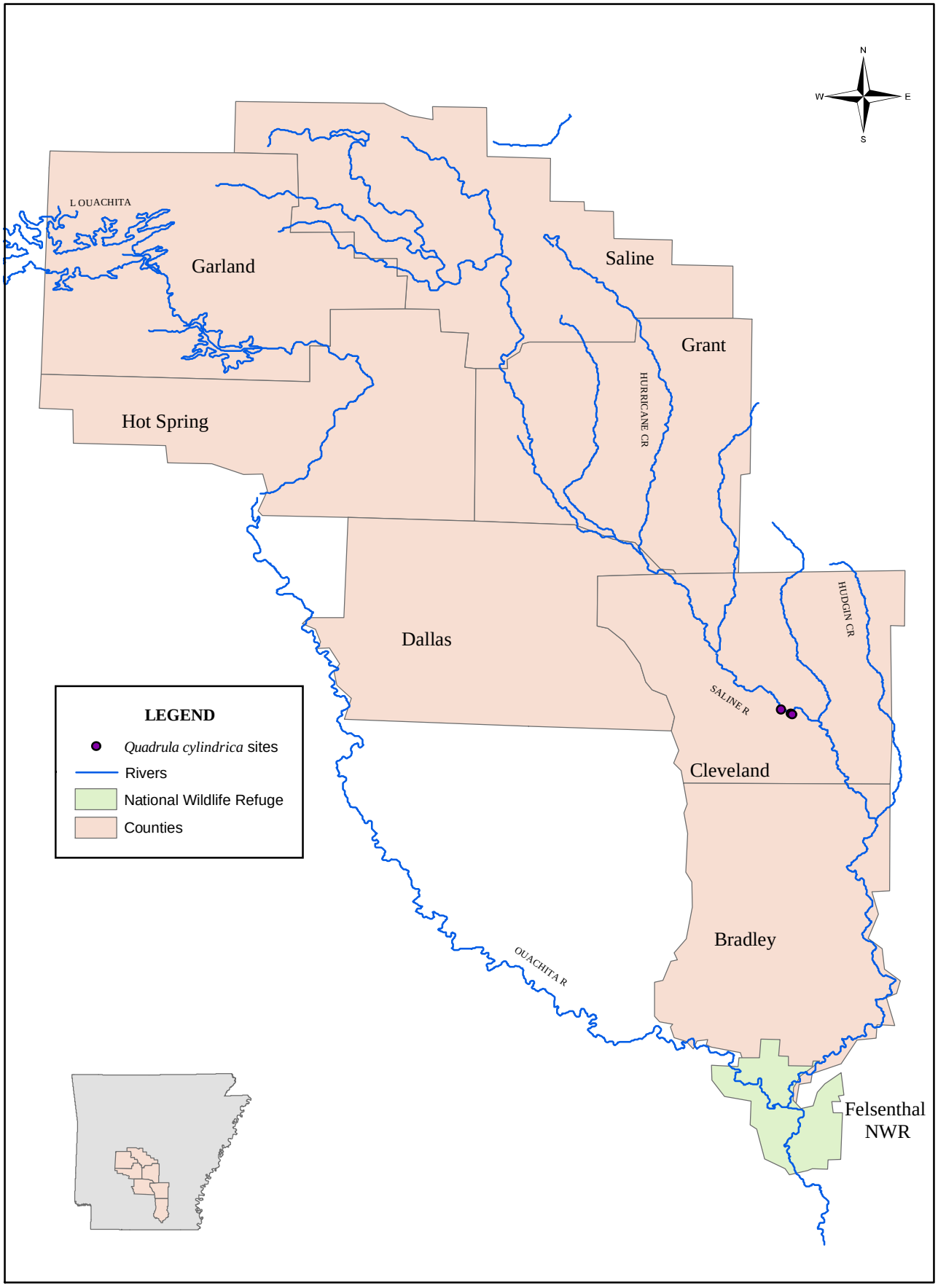


Figure 10. *Quadrula cylindrica* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

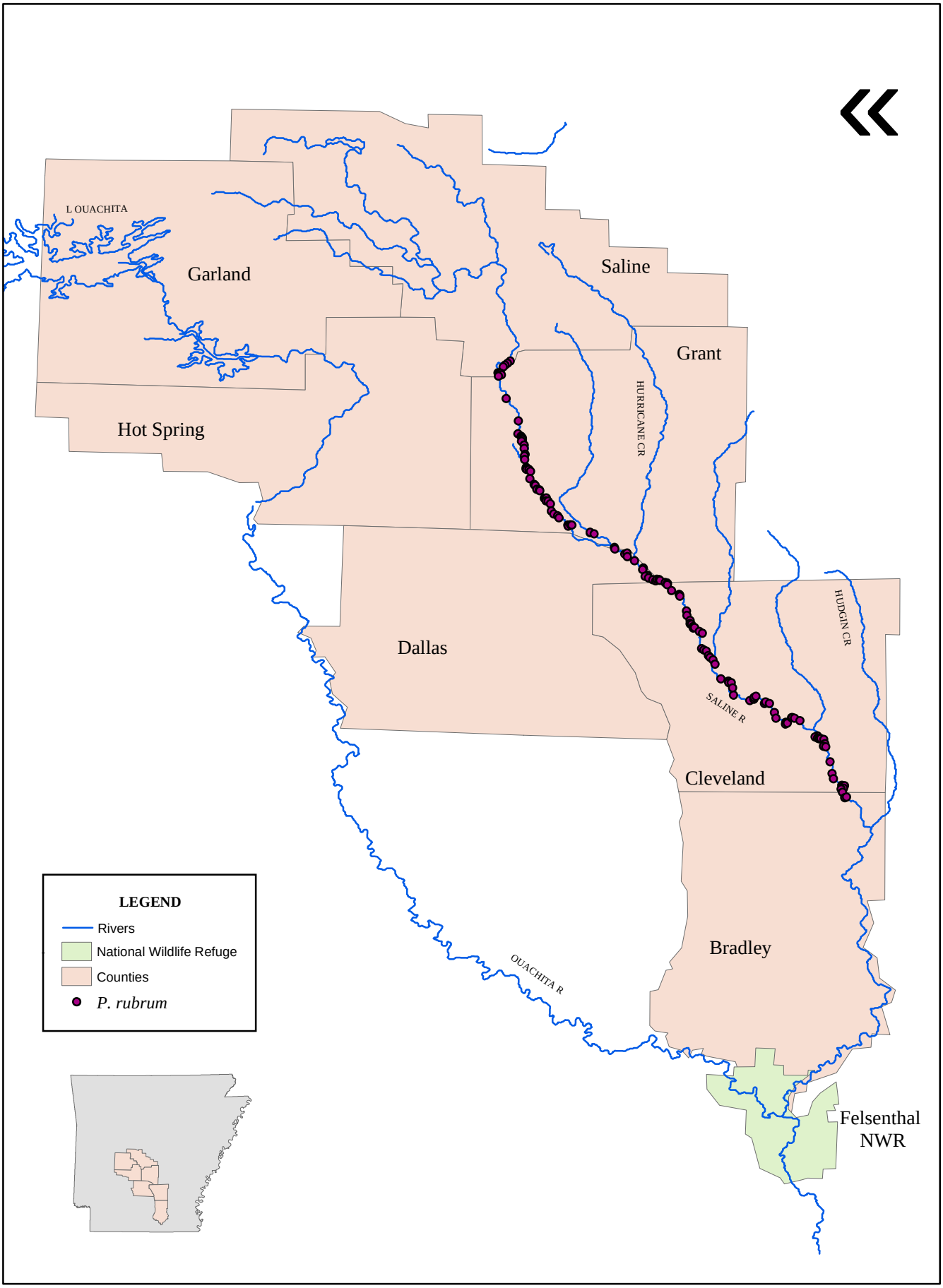


Figure 11. *Pleurobema rubrum* sites in the Saline River from near Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas, 2001 - 2002.

Potential Factors Influencing Distribution

The intent of this section is to provide a description of physical disturbance(s) in the Saline River that may be influencing mussel occurrence/distribution. The reach from Arkansas Game and Fish Commission's (AGFC) Fitzhugh Landing to the Williams property, approximately 3.2 km (2 rm) upstream of Arkansas Highway 270, consisted of numerous logjams and unstable bank sections. Very few live mussels or relicts were encountered in this reach. The Saline River also was "choked" by numerous logjams between SRB8 and Jenkins Ferry State Park (Plate 9). While this section was not devoid of mussel beds and concentrations, increased siltation existed in areas near logjams.

The appearance of the Saline River changes in the Big Island reach, a 11.3-km (7-mile) braided channel. The riparian zone consists of bottomland hardwoods and cypress. Natural channel constriction and divergence appears to have a "mouse trap" effect on large woody debris transported by high flow events. The result of this natural fluvial process is increased siltation during high flow conditions caused by logjams. This portion of the river is nearly impassable by canoe. Downstream of Big Island the channel narrows, depth remains two to eight feet, velocity appears to increase and the river takes on more of a large river appearance (fewer riffles). This is the first reach that mussel beds appear in river bends, rather than being confined to riffle/run series.



Plate 9. Saline River logjam.

Riffle frequency and areas of sand increase from Arkansas Highway 167 to the confluence of Hurricane Creek. The gradient lessens and depositional areas increase from White Oak Bluff to Mahoney's Ferry. A major logjam downstream of SRB38 blocks most boat passage downstream at low flow conditions.

Gum Springs area to SRC46 is dominated by depositional substrates. Few mussels were encountered in areas with gravel. Relative mussel bed size increases downstream from SRB52. Riffles become less frequent in this reach of the river, although the river generally remains shallow (3 to 8 feet at low flow conditions). Numerous logs and physical disturbance impacted a long series of shoals from SRC48 to Charnnings Ferry. Charnnings Ferry to SRB56 was depositional. These two reaches did not provide suitable habitat to support mussels.

The Saline River depth decreases downstream of Arkansas Highway 79 to SRC51, possibly a result of increased physical disturbance from bank erosion and tree fall. Increased sedimentation was observed in this reach. Mussel beds were associated with riffle/run series similar to beds located upstream of Arkansas Highway 167, except water depth was one to three feet deeper. Riffle/run sequences are common from SRC51 to Mt. Elba. This reach resembles portions of the river upstream of Arkansas Highway 167,

with fewer logjams. Better stability, less physical disturbance, contributes to greater bed area.

SRB77 to Vince Bluff is shallow consisting of numerous riffle/run series. Beds are typically upstream of these areas and/or in river bends. Inside banks offer greater stability because there is an absence of bank sloughing. However, bank sloughing does not appear to be a major problem. Water depth and siltation increases downstream of Vince Bluff for approximately two to three river miles. The reach illustrated on the U.S.G.S. Warren NE topographic map consists of large gravel/sand bars, shallow water and numerous mussel beds associated with riffle/run series in river bends.

DISCUSSION

Taxonomy

Taxonomic uncertainties exist in the Arkansas mussel fauna for *Fusconaia*, *Pleurobema*, *Lampsilis*, *Lasmigona*, *Cyprogenia*, and *Obovaria* species. Recent and ongoing phylogenetic analyses that incorporate genealogical relationships of alleles have demonstrated success in helping to disentangle historical and biogeographical patterns in the genetic diversity of unionids. Side-by-side comparison of nested cladistic analysis and analysis of molecular variance indicated congruent support for a fragmentation event separating Arkansas River basin localities from all other *Lampsilis hydiana* localities (Turner et al. 2000). The oldest event was the separation of the Arkansas River population from the remainder of populations located farther southwest in the Ouachita and Saline river drainages. The second event was inferred between the upper Saline River and the remainder of the population in the lower Saline and Ouachita rivers. Turner et al. (2000) results support identifying specimens of the *Lampsilis hydiana* complex as *Lampsilis hydiana* in the Saline River.

Taxonomic questions concerning the genus *Cyprogenia* may have significant implications regarding the status of *C. aberti* and *C. stegaria. C. stegaria,* an endemic to the Ohio, Cumberland and Tennessee river drainages (Ortmann 1918, 1919) and Federally Endangered, is only known to have three reproducing populations (Parmalee and Bogan 1998). Serb and Obermeyer (2001) indicate that *C. aberti* may not be a monophyletic group in Arkansas. They suggest that populations of *Cyprogenia* have been geographically isolated from one another. Their results found that the Ouachita River basin population grouped together as two very distinct clades, *C. aberti* and a second clade that is more closely related to *C. stegaria* from the Clinch River (Tennessee) than to other *C. aberti* populations from Arkansas, Missouri and Kansas. Individuals from a single locality in the Saline River may be distributed across both clades. The genetic distance between the two clades is much higher than typically seen in intraspecific studies of invertebrate populations. Initially, this suggests there are two taxa in the Saline River. Additional sequence data will be necessary to determine if these two clades are "real" of if they are a result of mitotype sampling error. Analysis of nuclear DNA would answer this question. All *Cyprogenia* specimens collected during this study

are referred to as *C. aberti* because insufficient data exists at this time to separate the genus.

Researchers are currently conducting a genetic analysis of *Lasmigona*. No preliminary data from this study were available for this report. However, the possibility exist that two forms of *Lasmigona, L. costata* and *L. nuda,* may occur in the Ouachita River Basin. The form resembling *L. costata* was the only form encountered during this survey.

Malacologists at the University of Alabama are currently conducting genetic analyses on the *Fusconaia* and *Pleurobema* species occurring in the southeastern United States. John Harris (pers. comm.) believes that the *Fusconaia/Pleurobema* complex in Arkansas consists of four species, *P. sintoxia, P. rubrum, F. flava* and *F. ebena*. We recognized *F. flava, F. ebena* and *P. rubrum* during this survey. A fourth, and maybe fifth, *Fusconaia/Pleurobema* species may occur in the Saline River. Since insufficient data exists at this time to accurately separate this complex, we referred to the complex, excluding *F. flava, F. ebena* and *P. rubrum*, as *Fusconaia/Pleurobema* type.

Understanding how genetic variation is partitioned across a species geographic range is a critical first step to initiating recovery plans. If care is not taken to first identify this genetic partitioning, captive breeding and re-introducing species into parts of a species range may have adverse effects such as outbreeding depression, loss of co-adaptive gene complexes and loss of valuable natural variation (Avise 1989). Future genetic studies should resolve taxonomic questions surrounding fragmented populations in the Saline River.

Endangered, Threatened and Special Concern Species

Three federally listed endangered or threatened species, *Lampsilis powelli, Lampsilis abrupta* and *Quadrula fragosa*, were encountered during this survey from 22 new locations. Eight species, *Cyprogenia aberti, Ligumia recta, Lampsilis ornata, Obovaria jacksoniana, Pleurobema rubrum, Quadrula cylindrica, Toxolasma lividus* and *Villosa arkansasensis*, of global or state concern were encountered during this survey. Global and state concern species were common, and sometimes locally abundant, throughout the study area. A summary of each species distribution in the Saline River is provided in the results section.

Lampsilis powelli has historically been reported from the upper forks of the Saline River (Harris et al. 1997). The known range of this species was extended by 41.9 km (26 rm) to Arkansas Highway 270. *Lampsilis abrupta* is generally considered an inhabitant of medium to large rivers, and it is usually associated with mussel beds. This species prefers moderate to swift current over sand/gravel or gravel substrates. *Quadrula fragosa* is generally considered an inhabitant of medium to large rivers. The species prefers mud, sand and gravel substrates.

Management strategies should concentrate on maintaining the quantity and/or quality of habitat throughout the study area in order to sustain viable populations. Areas requiring special management attention are identified in the results section.

Distribution

Mussels exist in areas where substrate and environmental conditions permit, but these areas were small and less numerous upstream of Arkansas Highway 167 and subject to natural and/or anthropogenic disturbances. Reaches downstream of Arkansas Highway 167 generally had less disturbance and deeper water.

Gordon et al. (1979) reviewed literature and museum specimens on Arkansas' unionacean fauna and subsequently listed 35 species as known or possible elements of the Saline River unionacean fauna. Davidson (1997) confirmed the presence of 24 of 35 species suspected by Gordon et al. (1979) as elements of the lower Saline River fauna. Four additional taxa mentioned by Gordon et al. (1979) were confirmed during this survey. Four species, *Toxolasma parvus, T. glans* and *Ligumia subrostrata, Uniomerus tetralasmus*, reported by Gordon et al. (1979) have not been encountered in surveys of the mainstem Saline River, but may exist in oxbows and side channels where more favorable habitat occurs. Three species, *Pleurobema cordatum coccineum, P. c. plenum* and *Lampsilis siliquoidea*, listed by Gordon et al. (1979) are not cited by investigators in recent surveys of the Saline River as elements of the mussel fauna. Davidson (1997) and results included in this report list 12 species not reported by Gordon et al. (1979) as elements of the Saline River mussel fauna. Forty-two species are now known to occur in the Saline River, an additional 4 species may occur in backwater areas, and genetic analyses may add an additional 4 species.

The introduced Asian Clam *Corbicula fluminea* was found throughout the present study area. This species invaded nearly every major river system in the United States since its introduction sometime during or before the 1920's.

Davidson (1997) reported 10 mussel beds with a mean density of 28.5 mussels/m^2 from the lower 18 km of the Saline River. Davidson and Gosse (2001) discovered two mussel beds near the Holly Creek confluence, upstream of Tull, Arkansas. Ninety-five mussel beds were encountered during this survey, bringing the total number of mussel beds for the mainstem Saline River to 107 mussel beds, with 81.6 km (50.6 rm) unsurveyed. This greatly exceeds the total number of mussel beds reported for any other river in Arkansas. Posey (1997) and Harris (1999) discovered 66 mussel beds in the mainstem Ouachita River. Davidson (1997) compared mussel bed densities from large rivers in the Ouachita, St. Francis and White river drainage basins and reported that densities were greater in the Ouachita River Basin. These numbers suggest that the Ouachita River Basin supports a larger mussel community than other drainage basins in Arkansas.

Very little quantitative data exists for mussel beds in the Saline River. Davidson (1997) reported community estimates ranging from 24,500±13,600 to 237,486±42,385 mussels per bed in the lower Saline River. While Davidson and Gosse (2003) reported

community estimates ranging from 991±683 to 3,357±1,163 mussels per bed in the upper Saline River. Rust (1993), Christian (1995) and Posey (1997) noted general trends of increased mussel bed size, community standing crop and species diversity from upstream to downstream within their respective study areas which included the Black, Cache, Ouachita and White rivers in Arkansas. No quantitative data were collected during this survey. However, data indicate that standing crop of mussel beds will be lower in the upper Saline River because bed area and mussel density is less. Some larger beds in the lower Saline River may exceed 300,000 individuals. Species richness within mussel beds was generally greater in beds downstream of Canoe Landing, 6 to 18 species per bed ($\bar{x}$ = 11) upstream versus 8 to 21 species per bed ($\bar{x}$ = 14) downstream. Future efforts should concentrate on collecting quantitative data for mussel beds in order to monitor long-term population trends.

Rust (1993), Christian (1995) and Posey et al. (1996) reported shifts in species as one moves from upper reaches towards the mouth of a river. Longitudinal zonation in species was noted in the Saline River. Burns and McDonnell (1992a) reported that *Amblema plicata* and *Lampsilis hydiana* are the most commonly encountered species in the forks of the Saline River. *Actinonaias ligamentina* is dominant in the upper reaches of the Saline River from the convergence of the four forks to the vicinity near Arkansas Highway 270. Between Arkansas Highway 270 and Jenkins Ferry State Park a shift in dominance occurs that favors *Fusconaia/Pleurobema* type and *Pleurobema rubrum*. Downstream of Arkansas Highway 270 *Elliptio dilatata* is the dominant species encountered until SRB80 where *A. ligamentina* makes a resurgence. Davidson (1997) reported *A. plicata* and *Quadrula pustulosa* as the dominant species in the 18 km upstream of the mouth. Hydrologic variability is important in determining fish distribution (Horwitz 1978) and mussel distribution (Strayer 1983). These two factors, hydrologic variability and fish host distribution/abundance, may best explain the longitudinal zonation in species that occurs in the Saline River. A more detailed analyses of multiple river systems should help malacologists better understand the mechanisms affecting mussel distribution.

Additional Management Concerns and Needs

Through communication with local residents, the authors learned about a large land sale (approximately 9,000 acres) and subsequent timber harvest planned for land parcels, owned by International Paper Company, bordering the Saline River between Arkansas Highwy 46 and Arkansas Highway 167. Riparian forests have many important roles in the structure and function of ecosystems. Loss of streamside forest typically results in accelerated bank erosion, channel widening and shallowing, increases in stream temperature, loss of aquatic and riparian habitat and other effects. Conservation agencies should work with stakeholders to ensure that streamside protection zones are adequate to protect "beneficial uses" of stream resources. Improperly maintained streamside protection zones may adversely affect species diversity and biological productivity by degrading water quality (i.e. nutrient loading, sediment loading), energy sources and altering flow regimes and physical habitat (Roell 1994).

One of the qualities of the Saline River is the relatively unmanaged aspect of the large tree jams that are well recorded in pre-settlement literature. For reasons that are not easily elucidated, the large tree jams are part of critical ecosystem processes that probably maintain other aspects of biodiversity and system integrity. However, the tree jams do not favor stable mussel habitat in the immediate vicinity. "Snag and drag" operations have been proposed in the Saline River. These operations may alter ecosystem processes by removing tree jams thereby altering critical mussel habitat and potentially extirpating federally threatened and endangered species and species of global and state concern from the Saline River.

Protection or improvements in the status of mussels in the Saline River require proper management of the watershed and cooperative efforts of stakeholders. Private landowners and state and federal agencies should consider efforts to stabilize existing habitats (i.e. stream bank stabilization projects) and sustain exceptional water quality. Execution of these management strategies should improve and/or sustain the quality and quantity of vegetative cover in riparian zones, decrease siltation and subsequently improve habitat quality for mussels.

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APPENDIX 1

Locational data for mussel beds and concentrations in the Saline River
From Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas

Appendix 1. Locational data for mussel beds and concentrations in the Saline River from Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas. (Page 1 of 5)

	Latitude				Longitude							
Site ID	**Degrees**	**Minutes**	**Seconds**	**Decimal**	**Degrees**	**Minutes**	**Seconds**	**Decimal**	**Quad**	**Section**	**Township**	**Range**
SRC1	34	26	18.26	34.438406	92	36	7.52	-92.602089	Tull	NE1/4 NE1/4 27	3S	15W
SRC2	34	26	4.43	34.434564	92	36	23.41	-92.606503	Tull	NE1/4 SW1/4 27	3S	15W
SRC3	34	25	53.96	34.431656	92	36	38.88	-92.610800	Tull	NW1/4 SW1/4 27	3S	15W
SRC4	34	25	42.65	34.428514	92	36	49.56	-92.613767	Tull	SE1/4 SE1/4 28	3S	15W
SRC5	34	25	7.26	34.418683	92	37	21.68	-92.622689	Tull	NE1/4 SW1/4 33	3S	15W
SRB1	34	25	4.09	34.417803	92	37	14.45	-92.620681	Tull	NW1/4 SE1/4 33	3S	15W
SRB2	34	25	4.35	34.417875	92	37	6.33	-92.618425	Tull	NW1/4 SE1/4 33	3S	15W
SRC6	34	24	54.46	34.415128	92	37	0.16	-92.616711	Tull	SE1/4 SE1/4 33	3S	15W
SRC7	34	24	46.22	34.412839	92	37	18.67	-92.621853	Tull	NW1/4 NE1/4 4	4S	15W
SRC8	34	23	38.22	34.393950	92	37	0.79	-92.616886	Tull	NE1/4 NE1/4 8	4S	15W
SRB3	34	22	30.81	34.375225	92	36	32.43	-92.609008	Tull	NE1/4 SW1/4 15	4S	15W
SRC9	34	21	8.75	34.352431	92	35	37.09	-92.593636	Prattsville	NW1/4 NW1/4 26	4S	15W
SRC10	34	20	23.72	34.339922	92	35	12.05	-92.586681	Prattsville	SE1/4 SE1/4 26	4S	15W
SRB4	34	20	15.08	34.337522	92	35	18.27	-92.588408	Prattsville	NW1/4 NE1/4 35	4S	15W
SRB5	34	18	56.96	34.315822	92	35	20.22	-92.588950	Prattsville	NW1/4 SE1/4 2	5S	15W
SRC11	34	18	55.10	34.315306	92	35	12.86	-92.586906	Prattsville	NW1/4 SE1/4 2	5S	15W
SRC12	34	18	43.16	34.311989	92	35	7.52	-92.585422	Prattsville	SE1/4 SE1/4 12	5S	15W
SRB6	34	18	41.61	34.311558	92	35	1.39	-92.583719	Prattsville	SE1/4 SE1/4 2	5S	15W
SRB7	34	18	37.02	34.310283	92	34	58.72	-92.582978	Prattsville	NE1/4 NE1/4 11	5S	15W
SRC13	34	18	31.26	34.308683	92	34	53.85	-92.581625	Prattsville	NW1/4 NW1/4 12	5S	15W
SRC14	34	18	21.90	34.306083	92	34	53.90	-92.581639	Prattsville	SW1/4 NW1/4 12	5S	15W
SRB8	34	18	11.97	34.303325	92	34	59.13	-92.583092	Prattsville	SE1/4 NE1/4 11	5S	15W
SRC15	34	17	48.60	34.296833	92	34	42.79	-92.578553	Prattsville	SW1/4 SW1/4 12	5S	15W
SRC16	34	17	28.62	34.291283	92	34	42.35	-92.578431	Prattsville	SE1/4 NW1/4 13	5S	15W
SRC17	34	17	4.22	34.284506	92	34	29.05	-92.574736	Prattsville	NW1/4 SE1/4 13	5S	15W
SRB9	34	16	52.29	34.281192	92	34	36.63	-92.576842	Prattsville	NE1/4 NW1/4 24	5S	15W
SRC18	34	16	42.21	34.278392	92	34	41.15	-92.578097	Prattsville	NE1/4 NW1/4 24	5S	15W
SRC19	34	16	21.49	34.272636	92	34	39.82	-92.577728	Prattsville	NE1/4 SE1/4 24	5S	15W
SRC20	34	15	57.49	34.265969	92	34	36.72	-92.576867	Prattsville	NE1/4 NW1/4 25	5S	15W
SRB10	34	15	32.78	34.259106	92	34	29.94	-92.574983	Prattsville	NE1/4 SE1/4 25	5S	15W
SRB11	34	15	25.22	34.257006	92	34	30.92	-92.575256	Prattsville	NE1/4 SW1/4 25	5S	15W
SRC21	34	15	20.99	34.255831	92	34	26.92	-92.574144	Prattsville	SW1/4 SE1/4 25	5S	15W
SRC22	34	15	20.94	34.255817	92	34	13.27	-92.570353	Prattsville	SE1/4 SE1/4 25	5S	15W
SRB12	34	15	9.35	34.252597	92	34	4.36	-92.567878	Prattsville	SE1/4 SE1/4 25	5S	15W
SRC23	34	14	24.91	34.240253	92	34	8.39	-92.568997	Leola	SE1/4 SE1/4 36	5S	15W

Appendix 1. Locational data for mussel beds and concentrations in the Saline River from Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas. (Page 2 of 5)

	Latitude				Longitude							
Site ID	**Degrees**	**Minutes**	**Seconds**	**Decimal**	**Degrees**	**Minutes**	**Seconds**	**Decimal**	**Quad**	**Section**	**Township**	**Range**
SRB13	34	13	49.85	34.230514	92	33	41.95	-92.561653	Leola	SE1/4 NW1/4 6	6S	14W
SRB14	34	13	43.30	34.228694	92	33	35.88	-92.559967	Leola	SW1/4 NE1/4 6	6S	14W
SRC24	34	13	20.44	34.222344	92	33	26.40	-92.557333	Leola	SW1/4 SE1/4 6	6S	14W
SRB15	34	13	17.85	34.221625	92	33	10.93	-92.553036	Leola	SE1/4 SE1/4 6	6S	14W
SRC25	34	13	12.04	34.220011	92	33	8.56	-92.552378	Leola	SE1/4 SE1/4 6	6S	14W
SRB16	34	12	26.83	34.207453	92	32	41.22	-92.544783	Leola	SE1/4 SW1/4 8	6S	14W
SRC26	34	12	24.67	34.206853	92	32	33.09	-92.542525	Leola	SW1/4 SE1/4 8	6S	14W
SRB17	34	12	28.52	34.207922	92	32	27.72	-92.541033	Leola	SW1/4 SE1/4 8	6S	14W
SRB18	34	12	9.97	34.202769	92	32	29.92	-92.541644	Leola	NW1/4 NE1/4 17	6S	14W
SRC27	34	12	9.35	34.202597	92	32	17.13	-92.538092	Leola	NE1/4 NE1/4 17	6S	14W
SRC28	34	11	52.10	34.197806	92	32	3.73	-92.534369	Leola	SW1/4 NW1/4 16	6S	14W
SRB19	34	11	8.08	34.185578	92	31	57.45	-92.532625	Leola	NW1/4 NW1/4 21	6S	14W
SRB20	34	11	9.79	34.186053	92	31	36.99	-92.526942	Leola	NE1/4 NW1/4 21	6S	14W
SRB21	34	10	51.24	34.180900	92	31	42.81	-92.528558	Leola	NE1/4 SW1/4 21	6S	14W
SRB22	34	10	39.74	34.177706	92	31	19.63	-92.522119	Leola	SE1/4 SE1/4 21	6S	14W
SRC29	34	10	27.24	34.174233	92	31	12.22	-92.520061	Leola	NE1/4 NE1/4 28	6S	14W
SRC30	34	9	43.98	34.162217	92	30	18.40	-92.505111	Leola	SE1/4 SE1/4 27	6S	14W
SRB23	34	9	36.62	34.160172	92	30	15.98	-92.504439	Leola	SE1/4 SE1/4 27	6S	14W
SRB24	34	9	44.73	34.162425	92	30	1.51	-92.500419	Leola	SW1/4 SW1/4 26	6S	14W
SRC31	34	9	44.20	34.162278	92	29	53.91	-92.498308	Millerville	SW1/4 SW1/4 26	6S	14W
SRC32	34	8	57.62	34.149339	92	28	2.59	-92.467386	Millerville	NW1/4 SW1/4 31	6S	13W
SRB25	34	8	59.85	34.149958	92	28	1.91	-92.467197	Millerville	NW1/4 SW1/4 31	6S	13W
SRB26	34	8	49.89	34.147192	92	27	38.04	-92.460567	Millerville	SE1/4 SW1/4 31	6S	13W
SRC33	34	8	39.14	34.144206	92	26	55.39	-92.448719	Millerville	SE1/4 SE1/4 31	6S	13W
SRC34	34	7	28.64	34.124622	92	25	34.68	-92.426300	Bunn	SE1/4 NW1/4 9	7S	13W
SRB27	34	7	18.60	34.121833	92	25	31.52	-92.425422	Bunn	NE1/4 SW1/4 9	7S	13W
SRB28	34	6	48.60	34.113500	92	24	31.84	-92.408844	Bunn	NE1/4 NW1/4 15	7S	13W
SRB29	34	6	53.68	34.114911	92	24	18.07	-92.405019	Bunn	NW1/4 NE1/4 15	7S	13W
SRC35	34	6	32.47	34.109019	92	24	17.02	-92.404728	Bunn	SW1/4 NE1/4 15	7S	13W
SRC36	34	6	7.19	34.101997	92	23	32.19	-92.392275	Bunn	SE1/4 SW1/4 14	7S	13W
SRC37	34	6	8.01	34.102225	92	23	25.69	-92.390469	Bunn	SE1/4 SW1/4 14	7S	13W
SRC38	34	5	22.94	34.089706	92	22	39.81	-92.377725	Bunn	SW1/4 SW1/4 24	7S	13W
SRC39	34	5	14.70	34.087417	92	22	43.64	-92.378789	Bunn	NW1/4 NW1/4 25	7S	13W
SRB30	34	4	38.10	34.077250	92	22	13.20	-92.370333	Staves	SW1/4 SE1/4 25	7S	13W
SRB31	34	4	33.01	34.075836	92	22	28.48	-92.374578	Staves	SE1/4 SW1/4 25	7S	13W

Appendix 1. Locational data for mussel beds and concentrations in the Saline River from Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas. (Page 3 of 5)

	Latitude				Longitude							
Site ID	**Degrees**	**Minutes**	**Seconds**	**Decimal**	**Degrees**	**Minutes**	**Seconds**	**Decimal**	**Quad**	**Section**	**Township**	**Range**
SRB32	34	4	23.98	34.073328	92	22	8.41	-92.369003	Staves	SW1/4 SE1/4 25	7S	13W
SRB33	34	4	11.78	34.069939	92	21	41.60	-92.361556	Staves	NW1/4 NW1/4 31	7S	12W
SRC40	34	4	5.43	34.068175	92	21	23.96	-92.356656	Staves	SW1/4 NE1/4 31	7S	12W
SRB34	34	4	15.26	34.070906	92	21	16.92	-92.354700	Staves	31/32	7S	12W
SRB35	34	4	13.79	34.070497	92	21	5.27	-92.351464	Staves	NW1/4 NW1/4 32	7S	12W
SRB36	34	4	7.96	34.068878	92	20	58.16	-92.349489	Staves	NW1/4 NW1/4 32	7S	12W
SRB37	34	3	54.95	34.065264	92	20	26.28	-92.340633	Staves	NW1/4 SE1/4 32	7S	12W
SRC41	34	3	50.18	34.063939	92	20	13.15	-92.336986	Staves	32/33	7S	12W
SRB38	34	3	42.19	34.061719	92	20	13.77	-92.337158	Staves	SE1/4 SE14 32	7S	12W
SRC42	34	3	6.03	34.051675	92	19	48.28	-92.330078	Staves	NE1/4 SW1/4 4	8S	12W
SRB39	34	2	43.22	34.045339	92	19	1.24	-92.317011	Staves	NW1/4 NW1/4 10	8S	12W
SRB40	34	2	40.40	34.044556	92	18	59.38	-92.316494	Staves	NW1/4 NW1/4 10	8S	12W
SRC43	34	2	31.93	34.042203	92	18	57.57	-92.315992	Staves	SW1/4 NW1/4 10	8S	12W
SRC44	34	2	28.08	34.041133	92	18	41.43	-92.311508	Staves	SW1/4 NE1/4 10	8S	12W
SRB41	34	1	3.10	34.017528	92	18	18.04	-92.305011	Staves	15/22	8S	12W
SRB42	34	0	35.52	34.009867	92	18	14.69	-92.304081	Staves	NE1/4 SE1/4 22	8S	12W
SRB43	34	0	5.86	34.001628	92	17	56.30	-92.298972	Staves	NW1/4 NW1/4 26	8S	12W
SRB44	33	59	46.69	33.996303	92	17	55.77	-92.298825	Fordyce NE	SW1/4 NW1/4 26	8S	12W
SRB45	33	59	40.66	33.994628	92	17	47.27	-92.296464	Fordyce NE	NE1/4 SW1/4 26	8S	12W
SRB46	33	59	28.93	33.991369	92	17	38.47	-92.294019	Fordyce NE	SE1/4 SW1/4 26	8S	12W
SRB47	33	59	20.66	33.989072	92	17	36.15	-92.293375	Fordyce NE	NW1/4 NE1/4 35	8S	12W
SRC45	33	59	24.04	33.990011	92	17	28.60	-92.291278	Fordyce NE	SW1/4 SEE1/4 26	8S	12W
SRB48	33	59	0.23	33.983397	92	17	0.08	-92.283356	Fordyce NE	SW1/4 NW1/4 36	8S	12W
SRB49	33	58	49.01	33.980281	92	16	40.00	-92.277778	Fordyce NE	NE1/4 SW1/4 36	8S	12W
SRC46	33	57	16.00	33.954444	92	16	46.53	-92.279592	Fordyce NE	SW1/4 SW1/4 6	9S	12W
SRB50	33	57	8.11	33.952253	92	16	32.72	-92.275756	Fordyce NE	NW1/4 NE1/4 7	9S	11W
SRC47	33	57	4.90	33.951361	92	16	25.00	-92.273611	Fordyce NE	NE1/4 NE1/4 7	9S	11W
SRB51	33	57	0.51	33.950142	92	16	17.83	-92.271619	Fordyce NE	NE1/4 NE1/4 7	9S	11W
SRB52	33	56	33.41	33.942614	92	16	5.45	-92.268181	Fordyce NE	NW1/4 SW1/4 8	9S	11W
SRB53	33	56	23.26	33.939794	92	15	53.64	-92.264900	Fordyce NE	SE1/4 SW1/4 8	9S	11W
SRB54	33	56	8.14	33.935594	92	15	36.12	-92.260033	Fordyce NE	NW1/4 NE1/4 17	9S	11W
SRB55	33	55	41.19	33.928108	92	15	23.74	-92.256594	Fordyce NE	SE1/4 SE1/4 17	9S	11W
SRC48	33	55	36.97	33.926936	92	15	22.16	-92.256156	Fordyce NE	SE1/4 SE1/4 17	9S	11W
SRB56	33	54	12.58	33.903494	92	14	48.31	-92.246753	Rison	SE1/4 NW1/4 28	9S	11W
SRB57	33	53	58.48	33.899578	92	14	3.75	-92.234375	Rison	NW1/4 SW1/4 27	9S	11W

Appendix 1. Locational data for mussel beds and concentrations in the Saline River from Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas. (Page 4 of 5)

	Latitude				Longitude							
Site ID	Degrees	Minutes	Seconds	Decimal	Degrees	Minutes	Seconds	Decimal	Quad	Section	Township	Range
SRC49	33	53	48.86	33.896906	92	14	0.79	-92.233553	Rison	SW1/4 SW1/4 27	9S	11W
SRC50	33	53	48.77	33.896881	92	13	45.39	-92.229275	Rison	SE1/4 SW1/4 27	9S	11W
SRB58	33	53	17.25	33.888125	92	13	37.71	-92.227142	Rison	SW1/4 NE1/4 34	9S	11W
SRB59	33	52	50.37	33.880658	92	13	50.48	-92.230689	Rison	SE1/4 SW1/4 34	9S	11W
SRB60	33	52	34.31	33.876197	92	13	32.01	-92.225558	Rison	SW1/4 NE1/4 3	10S	11W
SRC51	33	52	0.66	33.866850	92	11	53.11	-92.198086	New Edinburg	SW1/4 SW1/4 1	10S	11W
SRB61	33	52	9.41	33.869281	92	11	28.62	-92.191283	New Edinburg	NW1/4 SE1/4 1	10S	11W
SRB62	33	52	19.63	33.872119	92	11	26.42	-92.190672	New Edinburg	NW1/4 SE1/4 1	10S	11W
SRB63	33	52	26.85	33.874125	92	11	16.34	-92.187872	New Edinburg	SE1/4 NE1/4 1	10S	11W
SRB64	33	52	27.30	33.874250	92	11	3.56	-92.184322	New Edinburg	SW1/4 NW1/4 6	10S	11W
SRB65	33	51	43.64	33.862122	92	10	23.74	-92.173261	New Edinburg	NW1/4 NE1/4 7	10S	10W
SRB66	33	51	52.77	33.864658	92	10	17.83	-92.171619	New Edinburg	NW1/4 NE1/4 7	10S	10W
SRB67	33	51	43.03	33.861953	92	9	53.53	-92.164869	New Edinburg	NW1/4 NW1/4 8	10S	10W
SRB68	33	50	48.62	33.846839	92	9	23.36	-92.156489	New Edinburg	NW1/4 NE1/4 17	10S	10W
SRB69	33	50	14.83	33.837453	92	9	14.83	-92.154119	New Edinburg	SW1/4 NE1/4 17	10S	10W
SRB70	33	49	50.05	33.830569	92	8	13.83	-92.137175	New Edinburg	SW1/4 NE1/4 21	10S	10W
SRB71	33	49	38.34	33.827317	92	8	16.53	-92.137925	New Edinburg	NW1/4 SE1/4 21	10S	10W
SRB72	33	49	43.93	33.828869	92	8	5.82	-92.134950	New Edinburg	NE1/4 SE1/4 21	10S	10W
SRB73	33	50	17.45	33.838181	92	7	39.31	-92.127586	New Edinburg	SW1/4 SW1/4 15	10S	10W
SRB74	33	50	18.77	33.838547	92	7	35.26	-92.126461	New Edinburg	SW1/4 SW1/4 15	10S	10W
SRB75	33	50	13.67	33.837131	92	7	22.26	-92.122850	Herbine	SW1/4 SE1/4 15	10S	10W
SRB76	33	49	59.62	33.833228	92	6	49.70	-92.113806	Herbine	NW1/4 NW1/4 23	10S	10W
SRB77	33	48	27.99	33.807775	92	5	3.34	-92.084261	Herbine	S1/2 25	10S	10W
SRB78	33	48	20.01	33.805558	92	5	16.97	-92.088047	Herbine	NW1/4 NE1/4 36	10S	10W
SRB79	33	48	13.27	33.803686	92	5	2.64	-92.084067	Herbine	NE1/4 NE1/4 36	10S	10W
SRB80	33	48	10.32	33.802867	92	4	52.78	-92.081328	Herbine	NE1/4 NE1/4 36	10S	10W
SRB81	33	48	10.96	33.803044	92	4	43.52	-92.078756	Herbine	SW1/4 NW1/4 31	10S	9W
SRB82	33	48	5.44	33.801511	92	4	24.01	-92.073336	Herbine	SE1/4 NW1/4 31	10S	9W
SRC52	33	47	39.71	33.794364	92	4	17.41	-92.071503	Herbine	SE1/4 SW1/4 31	10S	9W
SRB83	33	47	24.70	33.790194	92	4	26.60	-92.074056	Herbine	NE1/4 NW1/4 6	11S	9W
SRB84	33	47	20.28	33.788967	92	4	11.60	-92.069889	Herbine	NW1/4 NE1/4 6	11S	9W
SRB85	33	45	49.78	33.763828	92	3	45.49	-92.062636	Herbine	SW1/4 SW1/4 8	11S	9W
SRB86	33	44	39.05	33.744181	92	3	32.63	-92.059064	Warren NE	SW1/4 NW1/4 20	11S	9W
SRB87	33	44	6.95	33.735264	92	3	24.87	-92.056908	Warren NE	SE1/4 SW1/4 20	11S	9W
SRB88	33	43	27.54	33.724317	92	2	33.94	-92.042761	Warren NE	NW1/4 SW1/4 28	11S	9W

Appendix 1. Locational data for mussel beds and concentrations in the Saline River from Tull, Arkansas to Arkansas Highway 15 near Warren, Arkansas. (Page 5 of 5)

	Latitude				Longitude							
Site ID	**Degrees**	**Minutes**	**Seconds**	**Decimal**	**Degrees**	**Minutes**	**Seconds**	**Decimal**	**Quad**	**Section**	**Township**	**Range**
SRB89	33	43	17.47	33.721519	92	2	31.84	-92.042178	Warren NE	SW1/4 SW1/4 28	11S	9W
SRB90	33	43	24.95	33.723597	92	2	17.26	-92.038128	Warren NE	NE1/4 SW1/4 28	11S	9W
SRB91	33	43	9.39	33.719275	92	2	22.84	-92.039678	Warren NE	SE1/4 SW1/4 28	11S	9W
SRB92	33	43	3.76	33.717711	92	2	38.87	-92.044131	Warren NE	NW1/4 NW1/4 33	11S	9W
SRB93	33	42	44.80	33.712444	92	2	30.24	-92.041733	Warren NE	SW1/4 NW1/4 33	11S	9W
SRB94	33	42	12.46	33.703461	92	2	16.60	-92.037944	Warren NE	NE1/4 NW1/4 4	12S	9W
SRB95	33	42	16.06	33.704461	92	2	5.01	-92.034725	Warren NE	SW1/4 SE1/4 33	11S	9W

APPENDIX 2

Mussel bed and concentration data

Appendix 2. Mussel bed and concentration data. (Page 1 of 8)

Scientific Name	SRC1	SRC2	SRC3	SRC4	SRC5	SRB1	SRB2	SRC6	SRC7	SRC8	SRB3	SRC9	SRC10	SRB4	SRB5	SRC11	SRC12	SRB6	SRB7	SRC13	SRC14	SRB8
Actinonaias ligamentina	9	11	9	3	4	9	25	9	6	6	3			18	21	3	21	95	12	15	13	18
Alasmidonta marginata																					1	
Amblema plicata	2		3		3	4	R		1	R				1		1	2				1	1
Cyprogenia aberti			1				R	R						3				2	1	2	3	2
Ellipsaria lineolata																						
Elliptio dilatata	1	4	R	2		4	35	5	3					28	5		4	18	1	10		10
Fusconaia ebena	2	1	3		2	2	12	2			20	2		3	1		2		2		1	
Fusconaia flava			4	2		11	45	7	4	1	10	3		5	10	2	1	11	21	25	8	8
Fusconaia/Pleurobema type																						
Lampsilis abrupta														1								
Lampsilis cardium							6	6	1	1		2		3	2	3	6	8	2	4	3	3
Lampsilis hydiana							R	R														
Lampsils powelli		1			1	1	11	R		1				1							1	
Lampsilis teres										R						R		1				1
Lasmigona costata															1		1					1
Leptodea fragilis																						
Ligumia recta							2	R		R					1						3	
Megalonaias nervosa	1		1																			
Obliquaria reflexa														1								
Obovaria jacksoniana						1		3														
Obovaria olivaria																						
Plectomerus dombeyanus																						
Pleurobema pyramidatum	1	2	2	1	1	3	7	1	1		9			4	5			3	11	1	1	12
Potamilus purpuratus			R			1																
Ptychobranchus occidentalis			1		1	4	4	5	2	4	1			17			4	12	5	15		12
Pyganodon grandis																						
Quadrula cylindrica																						
Quadrula fragosa																						
Quadrula metanevra		1				1	25	R						2				4	1	3		1
Quadrula nodulata																						
Quadrula pustulosa	1	2			1	8	16	4	2	2	5		1	6	5	2	2	10	4	8	6	6
Quadrula quadrula																						
Strophitus undulatus														1			1	1				
Toxolasma lividus										R			2			R						
Tritogonia verrucosa															1	1						1
Truncilla donaciformis																						
Truncilla truncata														1								1
Utterbackii imbecillis																						
Villosa arkansasensis																						
Villosa lienosa										R												
Number of Live Individuals	17	22	24	8	13	48	188	42	20	15	48	7	3	95	52	12	44	165	60	83	41	78
Number of relicts	2	NC	3	NC	50+	NC	12	250+	NC	25	NC	NC	40+	1	NC	5	NC	NC	NC	NC	NC	NC
Number of Species	7	7	10	4	7	12	14	14	8	11	6	3	2	16	10	8	10	11	10	9	11	14
Area (m^2)	ND	ND	ND	ND	ND	340	150	150	5	ND	35	ND	75	60	225	25	ND	225	70	ND	ND	100
Approximate Density (#/m^2)	WS	WS	4	WS	WS	12	15	WS	8	WS	10	WS	WS	10	10	WS	4	15	15	WS	WS	13

NC - Not collected
ND - Not determined
WS - Widespread

Appendix 2. Mussel bed and concentration data. (Page 2 of 8)

Scientific Name	SRC15	SRC16	SRC17	SRB9	SRC18	SRC19	SRC20	SRB10	SRB11	SRC21	SRC22	SRB12	SRC23	SRB13	SRB14	SRC24	SRB15	SRC25	SRB16
Actinonaias ligamentina	4	5		18	7	20	2	10	12	5	7	22	8	8	38	6	11	5	14
Alasmidonta marginata																			
Amblema plicata			1	1	1			1				2		1	1			1	2
Cyprogenia aberti					1	5		1	1	1		10	1	1	6		1	1	2
Ellipsaria lineolata																			
Elliptio dilatata	2	1	2	7	2	27		3	6	6	8	17	8	6	53	8	17	8	5
Fusconaia ebena			1					1				1	1			1			
Fusconaia flava	10	9	1	16	4	12	1		R		1			2	2		1		
Fusconaia/Pleurobema type								15	30	10	4	20	9	8	72	16	32	5	23
Lampsilis abrupta										1									
Lampsilis cardium		1	3	5		4		1	1						1	1			1
Lampsilis hydiana																			
Lampsils powelli							1												
Lampsilis teres			1	1															
Lasmigona costata																			
Leptodea fragilis																			
Ligumia recta				2	1									1	1				2
Megalonaias nervosa															1				
Obliquaria reflexa																			
Obovaria jacksoniana																			
Obovaria olivaria																			
Plectomerus dombeyanus																			
Pleurobema pyramidatum	6	8		13	5	6		5	40	3	9	30	2	7	48	3	6	2	7
Potamilus purpuratus			1																1
Ptychobranchus occidentalis	17	12	4	7	7	9	R		2			8	1	3	2	3	2	1	6
Pyganodon grandis																			
Quadrula cylindrica																			
Quadrula fragosa																			
Quadrula metanevra	1	4	2	4		2			2	1	1	7	1			1	3		1
Quadrula nodulata												1							
Quadrula pustulosa	3	3	6	5		5		3	9	9	1	6		2	4	3	2	1	9
Quadrula quadrula																			
Strophitus undulatus								1											1
Toxolasma lividus																			
Tritogonia verrucosa			1												1				1
Truncilla donaciformis												1							1
Truncilla truncata				1		1						1			1				
Utterbackii imbecillis																			
Villosa arkansasensis							1												
Villosa lienosa																			1
Number of Live Individuals	43	43	23	80	28	91	5	41	103	36	31	126	31	39	231	42	75	24	77
Number of relicts	NC	NC	NC	NC	NC	NC	1	50+	1	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC
Number of Species	7	8	11	12	8	10	5	10	10	8	7	13	8	10	14	9	9	7	16
Area (m^2)	ND	20	ND	150	ND	200	ND	210	60	ND	ND	480	ND	80	630	300	750	20	75
Approximate Density (#/m^2)	5	10	WS	10	WS	5	WS	20	10	WS	WS	40	WS	15	20	8	15	6	13

NC - Not collected
ND - Not determined
WS - Widespread

Appendix 2. Mussel bed and concentration data. (Page 3 of 8)

Scientific Name	SRC26	SRB17	SRB18	SRC27	SRC28	SRB19	SRB20	SRB21	SRB22	SRC29	SRC30	SRB23	SRB24	SRC31	SRC32	SRB25	SRB26	SRC33	SRC34
Actinonaias ligamentina	7	6	10	3	10	7	3	7	12	5	3	4	4	6	3	13	7	5	6
Alasmidonta marginata																			
Amblema plicata		7	4		15				1			1	1			2	2		4
Cyprogenia aberti	2	3	2	1		4			3					1		9			
Ellipsaria lineolata																			
Elliptio dilatata	5		4	6	15	12	10	9	6	2	4	4	14	15	3	9	6	5	2
Fusconaia ebena					5		1	1	3			1						1	
Fusconaia flava																			10
Fusconaia/Pleurobema type	12	30	27	12	50+	11	50+	7	7	7	5	22	5	9	13	25	6	13	9
Lampsilis abrupta																			
Lampsilis cardium				1		1		1	1				1			1		1	
Lampsilis hydiana																			
Lampsils powelli																			
Lampsilis teres																			
Lasmigona costata						1													
Leptodea fragilis											1					1			
Ligumia recta																			
Megalonaias nervosa																			
Obliquaria reflexa		1	1			1										1			R
Obovaria jacksoniana					3										1				
Obovaria olivaria																			
Plectomerus dombeyanus							75+					R							
Pleurobema pyramidatum	4	8	11	4	50+	3		3	5	3	3	18	10	3	5	13	11		10
Potamilus purpuratus			1													1			
Ptychobranchus occidentalis			5	3		1		3	1	1		2				3	2	1	
Pyganodon grandis																			
Quadrula cylindrica																			
Quadrula fragosa																			
Quadrula metanevra		1				1		3				1	1	1	1	3	1	1	1
Quadrula nodulata												R							
Quadrula pustulosa	3	7		4	20	3	10	2	4		3					4	3	3	2
Quadrula quadrula																			
Strophitus undulatus								2									3		
Toxolasma lividus																			
Tritogonia verrucosa			2	1												1	1		
Truncilla donaciformis																			
Truncilla truncata															1		1		
Utterbackii imbecillis																			
Villosa arkansasensis																			1
Villosa lienosa																			1
Number of Live Individuals	33	63	67	35	168	45	149	35	43	18	19	53	36	35	27	86	43	30	46
Number of relicts	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	2	NC	NC	NC	NC	NC	NC	1
Number of Species	6	8	10	9	8	11	6	10	10	5	6	10	7	6	7	14	11	8	11
Area (m^2)	50	125	120	600	1,050	60	135	80	105	20	ND	90	300	125	40	300	350	20	70
Approximate Density (#/m^2)	8	12	12	8	6	12	10	14	12	4	WS	25	30	WS	3	35	25	5	7

NC - Not collected
ND - Not determined
WS - Widespread

Appendix 2. Mussel bed and concentration data. (Page 4 of 8)

Scientific Name	SRB27	SRB28	SRB29	SRC35	SRC36	SRC37	SRC38	SRC39	SRB30	SRB31	SRB32	SRB33	SRC40	SRB34	SRB35	SRB36	SRB37	SRC41	SRB38
Actinonaias ligamentina	10	1	6	3	7	2	1	6	10	16	19	22	4	17	7	1	19	10	8
Alasmidonta marginata																			
Amblema plicata	10	2		1	4	4	4	1	15	12	5	8	2			2	5		2
Cyprogenia aberti	1	1	3	1		1			3	1		3	1	1	2		1		1
Ellipsaria lineolata	1								1		1								
Elliptio dilatata	20	8	18	8	16	11	2	1	15	29	27	35	10	16	15	5	10	15	23
Fusconaia ebena												1	2	10				1	
Fusconaia flava	50+	12	3	3	3	2		4	23	17	11	14	4	12	3		2		5
Fusconaia/Pleurobema type	50+	6	7	10	7	2	6	2	20	28	18	9	3	1	5	5	10	6	9
Lampsilis abrupta	1				1					1									
Lampsilis cardium	1	1					1					1				1			
Lampsilis hydiana																			
Lampsils powelli																			
Lampsilis teres			1									1					1		
Lasmigona costata																			
Leptodea fragilis											1								
Ligumia recta																	2		
Megalonaias nervosa		1								1	1								
Obliquaria reflexa										1							1		
Obovaria jacksoniana					1														
Obovaria olivaria																			
Plectomerus dombeyanus																			
Pleurobema pyramidatum	50+	65	6	1	5		5	1	29	9	13	14	1	13	11	3	1	1	8
Potamilus purpuratus			1			R											1		
Ptychobranchus occidentalis		3	1		1					2	2	2	1	3	3	1	3		1
Pyganodon grandis																			
Quadrula cylindrica																			
Quadrula fragosa																			
Quadrula metanevra	2	3	2	2	2	3			2		2	3			1	1	1	1	2
Quadrula nodulata								1											
Quadrula pustulosa	50+	4	5	5	4	6	5	2	13	13	6	9		2	1		3		4
Quadrula quadrula																			
Strophitus undulatus									1		1			1			1		1
Toxolasma lividus																	1		1
Tritogonia verrucosa		1		1								2	3		1				
Truncilla donaciformis																			
Truncilla truncata									1			4		1					
Utterbackii imbecillis																			
Villosa arkansasensis																			
Villosa lienosa																			
Number of Live Individuals	246	108	53	35	51	31	24	18	133	130	107	128	31	77	49	19	62	34	65
Number of relicts	NC	NC	NC	NC	NC	1	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC
Number of Species	12	13	11	10	11	9	7	8	12	12	13	15	10	11	10	8	16	6	12
Area (m^2)	190	360	400	75	200	ND	300	300	490	400	880	645	200	750	300	160	100	150	225
Approximate Density (#/m^2)	20	30	25	8	8	WS	WS	WS	25	40	30	30	WS	30	20	20	20	8	25

NC - Not collected
ND - Not determined
WS - Widespread

Appendix 2. Mussel bed and concentration data. (Page 5 of 8)

Scientific Name	SRC42	SRB39	SRB40	SRC43	SRC44	SRB41	SRB42	SRB43	SRB44	SRB45	SRB46	SRB47	SRC45	SRB48	SRB49	SRC46	SRB50	SRC47	SRB51
Actinonaias ligamentina	6	14	14	7	2	13	22	32	10	11	8	6	5	24	14	8	26	10	6
Alasmidonta marginata																			
Amblema plicata	2		6	1	1	3	4	1	2	3	4	2		2	2				
Cyprogenia aberti	1	1		1		7	2	1		1			2	8			4		
Ellipsaria lineolata										1				1					
Elliptio dilatata	3	9	24	5	14	35	8	43	47	35	18	31	13	34	22	9	50	16	36
Fusconaia ebena								1		2		1	1	2		1			
Fusconaia flava		1	5	6		15	7	14	3	12	24	1	9	13	10	7	4		4
Fusconaia/Pleurobema type	7	2	9	8	2	8	3	19		17	10	3	2	8	2	5	6		6
Lampsilis abrupta			1					1		1									
Lampsilis cardium					1	4			1			1		1			12	2	
Lampsilis hydiana							R												
Lampsils powelli																			
Lampsilis teres			1			1													
Lasmigona costata																			
Leptodea fragilis																			
Ligumia recta			1																
Megalonaias nervosa						1													
Obliquaria reflexa																			
Obovaria jacksoniana							1							1					
Obovaria olivaria																			
Plectomerus dombeyanus							R												
Pleurobema pyramidatum	1	2	1	4		20	6	20	4	11	16	4	4	25	2	5	12		8
Potamilus purpuratus																			
Ptychobranchus occidentalis		2	4	1		3	1	3	3	4		1	1	3		1	4		
Pyganodon grandis																			
Quadrula cylindrica																			
Quadrula fragosa																			
Quadrula metanevra	1		1	1	1	8	2	5	1	4	4	1		7	4	3	6		
Quadrula nodulata			8																
Quadrula pustulosa	4		4	2	1	5	3	4		7	6	2	1	4	12	3	8	2	2
Quadrula quadrula																			
Strophitus undulatus								2		1	2	1	1	1					
Toxolasma lividus																			
Tritogonia verrucosa	1				1		1							1					
Truncilla donaciformis														1					
Truncilla truncata								2			6			3	2				
Utterbackii imbecillis																			
Villosa arkansasensis																			
Villosa lienosa										1									
Number of Live Individuals	26	31	79	36	23	123	60	148	71	111	98	54	39	139	70	42	132	30	62
Number of relicts	NC	NC	NC	NC	NC	NC	2	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC
Number of Species	9	7	13	10	8	13	14	14	8	15	10	12	10	18	9	9	10	4	6
Area (m^2)	300	70	490	30	60	300	150	885	105	650	320	200		700	350	200	360	250	60
Approximate Density (#/m^2)	6	15	25	15	6	25	25	30	15	45	30	25	7	45	25	5	30	5	20

NC - Not collected
ND - Not determined
WS - Widespread

Appendix 2. Mussel bed and concentration data. (Page 6 of 8)

Scientific Name	SRB52	SRB53	SRB54	SRB55	SRC48	SRB56	SRB57	SRC49	SRC50	SRB58	SRB59	SRB60	SRC51	SRB61	SRB62	SRB63	SRB64	SRB65	SRB66
Actinonaias ligamentina	13	11	16	16		22	17	7	3	59	12	15	14	3	17	13	5	29	13
Alasmidonta marginata																			
Amblema plicata	7	4	3	8		5	10		4	5	2			1	1	1	3		2
Cyprogenia aberti	1		3			1	1			1		1	1	1	1	1		6	3
Ellipsaria lineolata																		1	
Elliptio dilatata	48	29	30	22	10	49	34	13	5	85	12	37	17	23	63	43	33	74	43
Fusconaia ebena		1					2								1			3	
Fusconaia flava	13	18	7	8	1	15	12	4	2	22	4	3		2	3	2	4	10	5
Fusconaia/Pleurobema type	9	8	9	4	1	11	7			5	4		1		2	2	1	10	3
Lampsilis abrupta								1											
Lampsilis cardium		1	1			1	2			3			1			1		8	1
Lampsilis hydiana																			1
Lampsils powelli																			
Lampsilis teres		1											1						
Lasmigona costata																			
Leptodea fragilis																			
Ligumia recta												1	1		1			1	
Megalonaias nervosa		1																	
Obliquaria reflexa		1		2															1
Obovaria jacksoniana			1			1				1							1		
Obovaria olivaria							1												
Plectomerus dombeyanus																			
Pleurobema pyramidatum	6	6	4	4		9	5	6	3	10		2	2	1	3	1		2	3
Potamilus purpuratus	1						1						1					3	1
Ptychobranchus occidentalis	1					1	5			4					2	1	1	3	
Pyganodon grandis																			
Quadrula cylindrica																			
Quadrula fragosa																			
Quadrula metanevra		3	1			3	2		1	4		1	2	1		1	2	6	4
Quadrula nodulata													1				1		
Quadrula pustulosa	5	7	8	4	2	4	6		3	6	6	4	1	1	1	1	1	4	3
Quadrula quadrula																			
Strophitus undulatus		2	1							1	2	1	1		1	1		1	
Toxolasma lividus																			
Tritogonia verrucosa		2								1	2		2		1			2	1
Truncilla donaciformis																			
Truncilla truncata			2			1	1			3	2						1		1
Utterbackii imbecillis																			
Villosa arkansasensis																			
Villosa lienosa																			
Number of Live Individuals	104	95	86	68	14	123	106	31	21	210	46	65	46	33	97	68	53	163	85
Number of relicts	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC
Number of Species	10	15	13	8	4	13	15	5	7	15	9	9	14	8	13	12	11	16	15
Area (m^2)	2400	2500	2240	255	150	3950	350	50	200	980	150	700	425	300	600	560	125	1750	250
Approximate Density (#/m^2)	40	40	30	15	WS	50	25	WS	8	40	15	15	12	15	30	15	18	45	40

NC - Not collected
ND - Not determined
WS - Widespread

Appendix 2. Mussel bed and concentration data. (Page 7 of 8)

Scientific Name	SRB67	SRB68	SRB69	SRB70	SRB71	SRB72	SRB73	SRB74	SRB75	SRB76	SRB77	SRB78	SRB79	SRB80	SRB81	SRB82	SRC52	SRB83	SRB84
Actinonaias ligamentina	13	22	8	17	42	16	31	19	11	14	29	45	15	20	48	35	13	47	15
Alasmidonta marginata										1									
Amblema plicata	2	7	2	20	14	7	1	1	1	6	12	3	2	4	6	17	4	4	1
Cyprogenia aberti	3	3		6	15	10	3	2			5	8	3	3	1	9	1	2	
Ellipsaria lineolata				1												3			
Elliptio dilatata	59	45	34	32	68	29	40	38	9	20	62	46	40	5	87	29	17	64	39
Fusconaia ebena	1			1				1							1	1			
Fusconaia flava	9	6	7	20	37	10	3	5	4	5	11	11	10	11	5	18	1	9	6
Fusconaia/Pleurobema type	1	2	5	1	13	11	2	4	2	4	3	7	1	1	4	9	3	12	2
Lampsilis abrupta								1				1				1			
Lampsilis cardium				1	4	1	1	1	3	1	2	2	2	2	1	1	1	1	
Lampsilis hydiana										1					R				
Lampsils powelli																			
Lampsilis teres								1					1				1		
Lasmigona costata							1				1						1		
Leptodea fragilis				R	1														
Ligumia recta			1								2	1				2		R	
Megalonaias nervosa	1			1					1									1	
Obliquaria reflexa				1	5		1				1	3	1	1					
Obovaria jacksoniana					1	1		1		1	2	2		2		3			
Obovaria olivaria																			
Plectomerus dombeyanus				1															
Pleurobema pyramidatum	4	5	1	2	7	7	1	1	2	1	5	5	4	1	2	4	1	3	1
Potamilus purpuratus						1													
Ptychobranchus occidentalis	2	3	1	3	3	5	3	1	1	1	5	8	1	1	5	12	2	7	1
Pyganodon grandis																			
Quadrula cylindrica			1	4		1													
Quadrula fragosa					1														
Quadrula metanevra	3	1		4	12	11	4		1	1	9	6		9	5	3	1	9	3
Quadrula nodulata	R																		
Quadrula pustulosa	3	1	4	9	12	5	1	5	4	6	16	13	10	11	13	23	2	10	3
Quadrula quadrula																1			
Strophitus undulatus		2			1					1		2	1	1	1				
Toxolasma lividus				R															
Tritogonia verrucosa		1	1	6	3	2				1	1	1	2	2	1	1			1
Truncilla donaciformis																			
Truncilla truncata		1		1	3	3				1				1	1	1		2	
Utterbackii imbecillis																			
Villosa arkansasensis																			
Villosa lienosa																			
Number of Live Individuals	101	99	65	131	242	120	92	81	39	65	166	164	93	75	181	173	48	171	72
Number of relicts	1	NC	NC	300	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	1	NC	NC	NC	NC
Number of Species	13	13	11	21	18	16	13	14	11	16	16	17	14	16	16	19	13	13	10
Area (m^2)	480	450	240	5400	1980	400	315	400	120	100	8000	3900	750	1320	1950	2760	25	855	225
Approximate Density (#/m^2)	30	45	30	55	40	40	20	50	15	25	50	65	30	35	40	40	12	30	30

NC - Not collected
ND - Not determined
WS - Widespread

Appendix 2. Mussel bed and concentration data. (Page 8 of 8)

Scientific Name	SRB85	SRB86	SRB87	SRB88	SRB89	SRB90	SRB91	SRB92	SRB93	SRB94	SRB95
Actinonaias ligamentina	37	55	62	40	26	23	17	54	35	41	35
Alasmidonta marginata						1		1	1	2	
Amblema plicata	13	6	9	2	4	1		3	5		1
Cyprogenia aberti	3	6	4	4	3		2	9	3	4	3
Ellipsaria lineolata			2					1			
Elliptio dilatata	43	45	31	55	19	39	90	28	31	22	47
Fusconaia ebena				1					1		
Fusconaia flava	19	9	3	1	4	4	2	17	7		3
Fusconaia/Pleurobema type	4	2	4	4	3	7	2	11	3	2	2
Lampsilis abrupta	2										
Lampsilis cardium	1	2	11	3		3		1	1	1	2
Lampsilis hydiana											
Lampsils powelli											
Lampsilis teres											
Lasmigona costata						1					
Leptodea fragilis											
Ligumia recta	2	2		5	1			1	1	1	
Megalonaias nervosa	1		1								
Obliquaria reflexa		1	2						1		
Obovaria jacksoniana			1	3	1	1					
Obovaria olivaria											
Plectomerus dombeyanus											
Pleurobema pyramidatum	17	10	5	2	2	4	8	4	11	2	2
Potamilus purpuratus		1									
Ptychobranchus occidentalis	4	8	5	6	2	4	1	9	6	9	5
Pyganodon grandis											
Quadrula cylindrica											
Quadrula fragosa											
Quadrula metanevra	3	5	5	9	3	2	7	10	8	4	4
Quadrula nodulata											1
Quadrula pustulosa	5	14	22	1	9	14	2	15	9	3	9
Quadrula quadrula	1										
Strophitus undulatus	1		2								
Toxolasma lividus											
Tritogonia verrucosa	3	3	1	2							
Truncilla donaciformis											
Truncilla truncata	1	2	2	1				2	1		
Utterbackii imbecillis											
Villosa arkansasensis											
Villosa lienosa											
Number of Live Individuals	160	171	172	139	77	104	131	166	124	91	114
Number of relicts	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC	NC
Number of Species	18	16	18	16	12	13	9	15	16	11	12
Area (m^2)	6500	900	2150	960	360	450	960	1660	2000	1200	800
Approximate Density (#/m^2)	50	25	20	15	15	30	35	30	25	35	20

NC - Not collected
ND - Not determined
WS - Widespread

ADDENDUM 1

U. S. Geological Survey topographic maps with Saline River mussel beds and concentrations marked and labeled.

(Addendum 1 was provided to The Nature Conservancy Arkansas Field Office and the Arkansas Game and Fish Commission Mussel Program)